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RESEARCH ARTICLE

Influence of Climate Drivers on Extreme Precipitation in Bangladesh: Spatiotemporal Patterns and Underlying Mechanisms

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ABSTRACT

This study investigates the spatiotemporal variability of extreme precipitation indices (EPIs) and their linkages with large-scale climate drivers in Bangladesh. Daily precipitation data from 20 meteorological stations for the period 1980–2017 were analysed using the Mann–Kendall trend test and Sen's slope estimator to detect trends and abrupt change points, and complemented by Mantel tests, Geo-Detector modelling, and partial wavelet coherence (PWC) analysis to examine climatic influences. Among the eight EPIs considered, consecutive dry days (CDD), annual maximum 5-day precipitation (RX5DAY), and the frequency of heavy rainfall days (R10 and R30) exhibited increasing trends, whereas consecutive wet days (CWD), the simple daily intensity index (SDII), annual total wet-day precipitation (PRCPTOT), and maximum 1-day precipitation (RX1DAY) showed declining tendencies. Mutation analyses indicated turning points in 1997 for CDD and in 2004 for R10 and R30, marking intensified dry spells and shifts in heavy rainfall occurrence, while declines in CWD, SDII, and RX1DAY originated around 1988. Spatially, precipitation frequency and intensity are highest in the northeastern and southeastern regions, particularly along the coastal belt. Mantel and Geo-Detector results identified the Indian Ocean Dipole (IOD), El NIÑO-Southern Oscillation (ENSO), Arctic Oscillation (AO), and SUNSPOT as key climatic drivers influencing monsoon dynamics and moisture transport from the Bay of Bengal. PWC analysis further revealed positive influences of ENSO on rainfall frequency and negative associations of the AO and SUNSPOT with precipitation intensity. These findings enhance understanding of hydroclimatic extremes in Bangladesh and provide a scientific basis for adaptive water resources management and flood risk mitigation.

1 | Introduction

Extreme precipitation is a global concern that strongly affects floods, landslides, and water-resources management, particularly under a changing climate (Mishu et al. 2025). For example, the July 2004 flood in Bangladesh, triggered by intense rainfall, inundated nearly 50% of the country's territory at its peak, submerging most major river basins, causing more than 600 fatalities, and displacing about 30 million people (EMDAT 2022). Understanding the spatiotemporal behaviour of extreme precipitation is therefore crucial for climate-change assessment, hydrological design, and disaster forecasting (Bhattacharjee et al. 2025). In a warming climate, enhanced atmospheric moisture can substantially influence the frequency and severity of heavy precipitation across a range of spatial and temporal scales (Alexander et al. 2006; Li et al. 2025). Consequently, examining

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the evolution of extreme precipitation is essential for water managers and policymakers seeking to mitigate climate-related hydrological risks.

The spatiotemporal patterns of extreme precipitation have attracted increasing attention because of their implications for human systems and ecosystems (Li et al. 2021; Huguenin et al. 2023). Global analyses based on gridded daily precipitation datasets reveal significant increases in the frequency and magnitude of extremes and annual maxima in many regions (Alexander et al. 2006). In subtropical monsoon regions like Bangladesh, however, variability and trends in precipitation are often more complex, as these areas are particularly sensitive to climate change (Islam et al. 2020). Local geographic controls, including proximity to the coastline, elevation, topography, and land-use/land-cover change, can further modulate regional patterns of precipitation extremes (Das 2021; Ezaz et al. 2022).

Bangladesh ranks among the world's most climate vulnerable countries owing to its low-lying deltaic setting, high population density, and strong dependence on agriculture and water resources (Rahman and Islam 2019). Extreme precipitation events, including intense monsoon rainfall, severe pre-monsoon storms and short-duration cloudbursts, have major socio-economic and environmental consequences (Mishu et al. 2025). The extreme weather events regularly lead to destructive river and flash floods, riverbank erosion, urban waterlogging, and landslides in hilly regions, causing loss of life, infrastructure damage, crop failure, and prolonged livelihood disruption. Recent studies indicate that the frequency, intensity, and spatial distribution of extreme precipitation over South Asia, and Bangladesh in particular, are likely to change under continued global warming (Ezaz et al. 2022; Imran et al. 2023; Fahad et al. 2024). Nevertheless, substantial gaps remain in our understanding of historical trends, variability, and drivers of extreme events at regional and national scales. Bangladesh's unique location at the confluence of the Ganges–Brahmaputra–Meghna River system makes it especially sensitive to monsoon dynamics and large-scale climate variability (Li et al. 2025). A systematic assessment of extreme precipitation over Bangladesh is therefore critical to improving early warning systems, guiding climate-resilient infrastructure and agricultural planning, and strengthening disaster-risk-reduction strategies.

A number of studies have investigated changing precipitation characteristics in Bangladesh (Shahid 2010; Rakib 2019; Islam et al. 2020). Shahid (2010) reported an increase in the frequency of rainy months and a decrease in dry months over most regions for the period 1958–2007. Rakib (2019) noted a non-significant increase in the frequency of heavy rainfall days and consecutive wet days (CWD), and a non-significant decrease in CDD in eastern Bangladesh over 1960–2000. Islam et al. (2020) identified an increasing trend in decadal rainfall at a rate of 3.74 mm per decade in the southwest and south-central regions during 1975–2017. While these studies provide valuable insights into evolving precipitation regimes, they largely focus on mean or seasonal characteristics or on a limited set of indices and regions, and do not offer a comprehensive, country-wide assessment of the spatiotemporal variability of extreme precipitation indices (EPIs) (Islam et al. 2021). A detailed analysis of EPIs across Bangladesh, together with their spatially varying climatic drivers, is therefore still lacking.

Methodologically, many hydrometeorological studies have employed the Mann–Kendall test, sequential Mann–Kendall test, and Mantel test to investigate precipitation trends (Praveen et al. 2020). Most of these efforts have focused on time-series characterisation and trend detection, whereas relatively few have explicitly examined the climatic drivers of extreme precipitation with a view to improving water-resources planning and disaster prediction (Rahman and Islam 2019). Traditional approaches can be limited in their ability to attribute changes in extremes to specific drivers, particularly in the presence of non-linear relationships and complex spatial heterogeneity. Recently, hydroclimatologists have increasingly adopted advanced statistical tools such as the Geo-Detector model and wavelet-based methods, including partial wavelet coherence (PWC), to disentangle the influence of large-scale climate modes on hydroclimatic variables (Ren et al. 2024). These approaches have shown promise in revealing both linear and non-linear linkages between climatic drivers and extreme weather events.

Attribution of changes in extreme precipitation remains challenging because multiple large-scale modes and local processes can act synergistically. Relevant climatic drivers include the El NIÑO-Southern Oscillation (ENSO) (An et al. 2023), Indian Ocean Dipole (IOD) (Ahmed et al. 2017), Arctic/Atlantic Oscillation (AO) (Ayala 2019), SUNSPOT (Rahman and Islam 2019), and local circulation and land-surface conditions (Ren et al. 2024). Global warming can enhance atmospheric moisture availability and thereby intensify extreme precipitation (An et al. 2023). Extensive work worldwide has shown that EPIs are often modulated by these large-scale 'teleconnection' patterns (Wahiduzzaman and Luo 2021; Ren et al. 2024). In Bangladesh, several studies have examined the link between climatic drivers and EPIs (Wahiduzzaman and Luo 2021; Ezaz et al. 2022; Imran et al. 2023). Wahiduzzaman and Luo (2021) and Rahman and Islam (2019) highlighted the influence of ENSO and SUNSPOT on precipitation extremes. Although ENSO, and Multidecadal Oscillation (AO) have been associated with extreme precipitation over Bangladesh, India, Nepal and Southeast Asia (Ezaz et al. 2022; Zhao et al. 2023; Chaulagain et al. 2025; Alizadeh and Mousavizadeh 2025), the combined and evolving effects of these drivers on EPIs across Bangladesh at the spatiotemporal scale remain poorly constrained. Addressing this gap is important for understanding precipitation variability under climate change and for improving predictive capability for hydrological extremes.

Extreme precipitation indices thus serve both as indicators and as consequences of climate change in Bangladesh. Observed shifts in the intensity, frequency, and spatial distribution of EPIs suggest that warming is altering monsoon dynamics and associated teleconnections. Accurate information on EPIs is crucial for managing hydrological extremes and designing long-term adaptation strategies. However, most EPI-based studies in Bangladesh have been limited to specific regions or a small number of stations. For example, Basher et al. (2018) analysed EPIs using data from only eight stations in northern Bangladesh, while Shirin et al. (2022) used six stations to identify changes in EPIs in the southwest, finding increased intensity, duration, and

frequency of extremes alongside declining total rainfall and increased CDD. These studies reveal spatially heterogeneous and sometimes inconclusive patterns, and the broader spatiotemporal structure of EPIs across the country remains insufficiently understood. Despite recent efforts (Rahman and Islam 2019; Abdullah et al. 2020; Imran et al. 2023), there is still a lack of systematic, nationwide analysis of EPIs and their climatic controls over multiple decades.

Motivated by these gaps, this study addresses the following research questions:

Q1. *What are the spatiotemporal trends of EPIs across Bangladesh over the last four decades?*

Q2. *Are there notable mutation (change) points in these EPIs, and when do they occur?*

Q3. *What are the key large-scale climatic drivers influencing EPIs in Bangladesh, and how do these relationships vary in space and time?*

Therefore, the objectives of this research are: (1) to characterise the spatiotemporal trends of EPIs across Bangladesh for the period 1980–2017 and to identify mutation years using the Mann–Kendall test, Sen's slope estimator and the Mann–Kendall mutation test; and (2) to investigate the links between EPIs and large-scale climatic drivers from linear, non-linear and time–frequency perspectives using Pearson correlation, Geo-Detector and PWC. The findings are expected to provide scientific support for climate adaptation and disaster-risk management in Bangladesh and to assist policymakers in formulating strategies to cope with climate change and mitigate the impacts of extreme precipitation.

2 | Data and Methods

2.1 | Study Area

Bangladesh is located in South Asia, bordering India and Myanmar, between 20°34′–26°38′N and 88°01′–92°41′E (Figure 1). The country covers approximately 147,570 km^2 and is dominated by low-lying floodplains, with hilly terrain confined mainly to the south-eastern and north-eastern regions (Rahman and Islam 2019). The climate is humid subtropical and strongly influenced by the south-west monsoon, with four distinct seasons: hot summer, monsoon, post-monsoon, and cool, dry winter. The combination of its extensive deltaic plains, highly variable climate, widespread poverty, and strong dependence on climate-sensitive sectors such as agriculture contributes to pronounced hydrometeorological and socio-economic vulnerability. Climatically, Bangladesh can be divided into seven regions: south-eastern, north-eastern, northern, north-western, south-western, western and south-central (Islam et al. 2020). The south-eastern region is characterised by heavy rainfall and relatively lower temperatures, whereas the north-eastern region experiences intense summer heat, high monsoonal rainfall and cloudy winters. The northern region is marked by high rainfall, hot summers and cool winters, while the north-western region is prone to very hot summers with comparatively modest rainfall. In the south-west, both rainfall and temperature show moderate spatial variability, and the south-central zone is distinguished by relatively mild summers and high rainfall (Imran et al. 2023). More than 80% of the country's annual rainfall occurs during the monsoon season, with a national mean of about 2428 mm (Islam et al. 2021). Monsoon depressions forming over the Bay of Bengal typically propagate northwards into the Indian subcontinent from July to September, exerting a strong influence on Bangladesh's rainfall regime. The mean annual temperature is approximately 25.7°C, with maximum temperatures around 31°C in August and minimum temperatures occurring in January up to 5.2°C (Islam et al. 2020).

2.2 | Data Sources and Quality Control

This study used daily precipitation records from 20 meteorological stations across Bangladesh (Figure 1) for the period 1980–2017, obtained from the Bangladesh Meteorological Department (BMD) (Islam et al. 2021). Although BMD operates 35 rainfall stations nationwide, their records differ substantially in duration, completeness and homogeneity, with observation histories ranging from 30 to 60 years (www.bmd.gov.bd). For long-term climate analysis, continuous and quality-assured datasets are essential. Consequently, 20 stations with the longest and most complete records were selected, following World Meteorological Organization (WMO) recommendations and criteria used in previous climatological studies in Bangladesh (Shahid 2010; Rahman and Islam 2019; Islam et al. 2020). These stations adequately represent the major hydro-climatic regions of the country, including coastal, floodplain, north-western drought-prone, north-eastern wet and hilly zones. Although more recent data exist for some meteorological sites, these have not yet undergone rigorous quality control and homogenisation; therefore, only verified datasets were used to ensure methodological consistency and to avoid biases associated with incomplete or unprocessed records (Table S1).

BMD routinely conducts preliminary screening, digitisation and basic homogenisation of long-term precipitation series, and these processed datasets formed the starting point for our analysis (Islam et al. 2021). All additional verification and quality-control procedures required for this study were undertaken independently. Approximately 5% of daily precipitation values were missing in the selected station records (Table S2). To preserve temporal continuity while minimising bias, missing values were infilled using a correlation- and distance-based normal-ratio method recommended by the WMO for climatological series (WMO 2024). For each target station, up to three neighbouring stations within 150 km were identified, provided that they exhibited strong Pearson correlation with the target series ($r \geq 0.8$) over the overlapping period. Short gaps (≤ 3 consecutive days) were filled using the daily normal-ratio method, in which missing precipitation is estimated as a weighted average of normalised precipitation from the neighbouring stations, with weights inversely proportional to distance. For longer gaps, the normal-ratio approach was applied at the monthly scale, and daily values were subsequently disaggregated in proportion to the neighbouring stations' daily rainfall. Days for which all neighbouring sites recorded zero rainfall were retained as zero-rain days to avoid introducing artificial extremes. Given

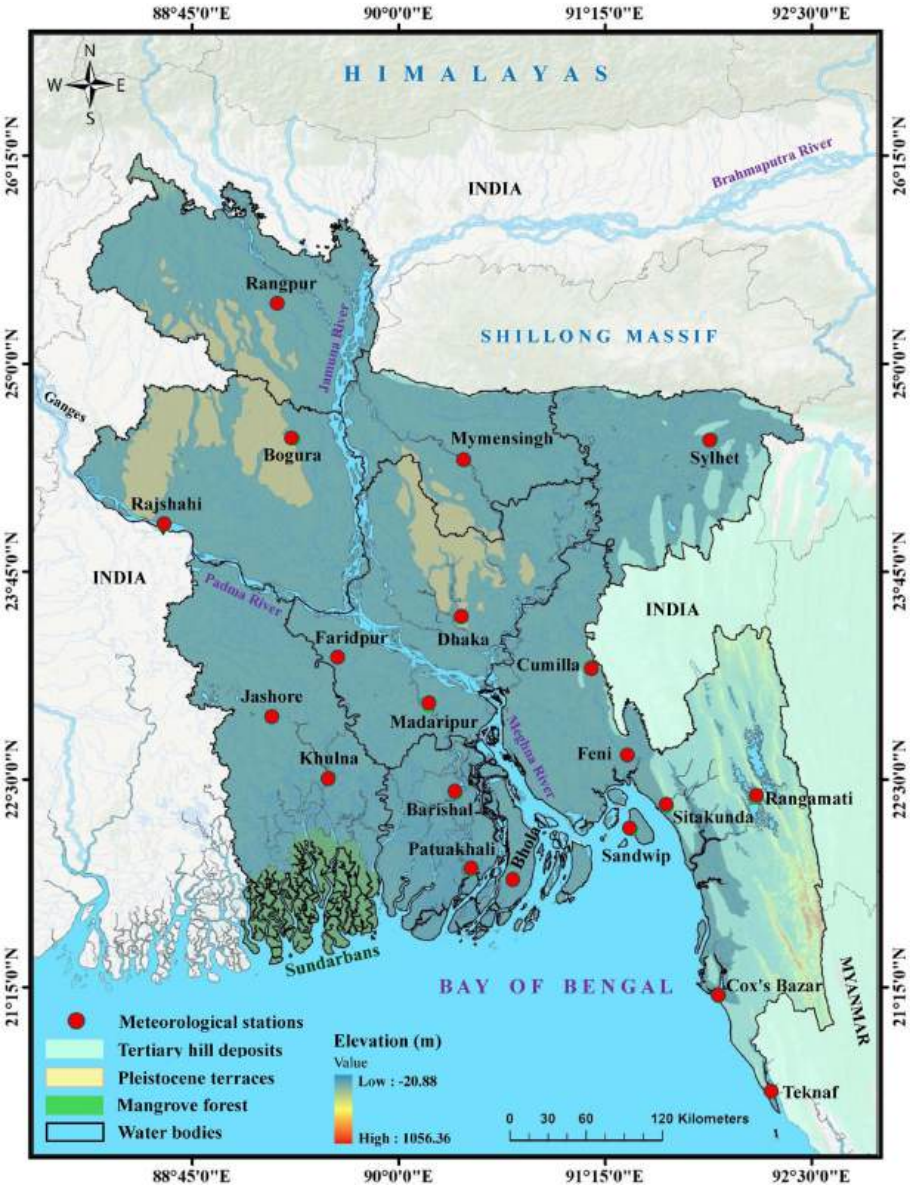

FIGURE 1 | Spatial distribution of meteorological stations and geographic landscape of Bangladesh (after Rashid et al. 2023). [Colour figure can be viewed at wileyonlinelibrary.com]

the small proportion of missing data and the use of only highly correlated neighbouring stations, the impact of infilling on both mean precipitation and extreme precipitation indices is expected to be minimal.

For the selected sites, a consistent quality-control and homogeneity assessment protocol was applied. Homogeneity was evaluated using the Standard Normal Homogeneity Test (SNHT). The test was applied individually to each station's annual precipitation time series, derived from the quality-controlled and infilled daily records. Annual precipitation totals were first computed from the daily data, and SNHT was then conducted on each station's annual series at a significance level of $p < 0.05$ to detect potential breakpoints associated with station relocation, instrumentation changes, or observational practice shifts. Homogeneity testing was performed on aggregated annual series rather than on raw daily data because daily precipitation records are highly variable and less suitable for detecting structural discontinuities. The results indicated no statistically significant breakpoints in the retained series; therefore, all 20 station records were considered homogeneous and suitable for subsequent trend and variability analyses. Serial correlation was examined at multiple lags, and cross-station correlation analysis was used to verify spatial consistency across climatic zones (Figure S1). These procedures ensured that all time series satisfied the requirements of continuity, homogeneity, and statistical robustness. Although the BMD datasets benefitted from prior institutional screening, all verification procedures directly underpinning the analyses reported here were performed independently as part of this research (Islam et al. 2021, 2023).

This study examines changes in the frequency and intensity of climate extremes using eight selected precipitation indices (Table 1). These indices are widely recognised as essential for detecting and characterising extreme climate and weather events (Alexander et al. 2006). Trends in each index were quantified

TABLE 1 | Definitions and classifications of the eight EPIs.

Index	Definition	Unit
Duration based indices		
CDD	Maximum sequence of dry days (precip < 1 mm)	days
CWD	Maximum sequence of wet days (precip ≥ 1 mm)	days
Intensity based indices		
RX1DAY	Highest 1-day precipitation total	mm
RX5DAY	Highest 5-day precipitation total	mm
SDII	Daily precipitation intensity index	mm/day
Threshold based indices		
R10	Number of days with precipitation ≥ 10 mm	days
R30	Number of days with precipitation ≥ 30 mm	days
PRCPTOT	Total annual wet-day precipitation (≥ 1 mm)	mm

using Sen's slope estimator, and their statistical significance was evaluated at the 5% level over the 37-year period from 1980 to 2017 (Imran et al. 2023).

To analyse the influence of large-scale climatic drivers on EPIs, monthly index data for the period 1980–2017 were obtained for the ENSO, IOD, Arctic Oscillation (AO), and SUNSPOT. ENSO was represented by the Niño3.4 sea surface temperature (SST) anomaly index (5°N–5°S, 170°–120°W), and IOD by the Dipole Mode Index (difference in SST anomalies between the western equatorial Indian Ocean (50°–70°E, 10°S–10°N) and the south-eastern equatorial Indian Ocean). AO was characterised by the normalised sea-level pressure anomaly index north of 20°N. Monthly SUNSPOT numbers were used as a proxy for external solar forcing. All indices were downloaded as quality-controlled monthly time series from the NOAA Climate Prediction Center and associated archives (http://www.cpc.ncep.noaa.gov/) and then truncated to the common analysis period 1980–2017.

To avoid ambiguity, all extreme precipitation indices in this study were derived exclusively from station-based daily precipitation records for 1980–2017. No gridded precipitation products were used at any stage. EPIs were computed directly at each station, and spatial fields were subsequently generated only for visualisation using Ordinary Kriging, without altering the underlying station-based values. Furthermore, although the Expert Team on Climate Change Detection and Indices (ETCCDI) framework commonly employs a 1971–2000 baseline for percentile-based indices, the present study uses only absolute- and duration-based indices (CDD, CWD, PRCPTOT, RX1DAY, RX5DAY, SDII, R10 and R30), none of which require a climatological baseline. Consequently, all indices were computed solely from the 1980–2017 station data, and no earlier BMD records or gridded baseline datasets were needed.

2.3 | Definition of EPIs

To characterise shifts in extreme precipitation, this research employed a set of eight EPIs recommended by the WMO and consistent with the Intergovernmental Panel on Climate Change (IPCC 2021; de Medeiros et al. 2022). All indices follow the standard definitions developed by the ETCCDI, a WMO–WCRP initiative that provides globally consistent metrics for detecting and comparing climate extremes (Ezaz et al. 2022). Although these indices are internationally standardised, previous studies have often adapted their application to regional hydroclimatic conditions (Islam et al. 2021).

The effective widely used approaches to identifying extreme climate events are applied such standardised indices to long-term observational records (Wang et al. 2022). In this study, eight EPIs were selected to examine spatial and temporal variations in extremes across Bangladesh for the period 1980–2017. Following Ren et al. (2024), these indices can be grouped into three functional categories: first, precipitation frequency class indices (PFCIs) include CDD (consecutive dry days), CWD, R10 (number of days with precipitation ≥ 10 mm; heavy precipitation days) and R30 (number of days with precipitation ≥ 30 mm; very heavy precipitation days), which describe the frequency and persistence of dry and wet spells. Second, extreme weather intensity and persistence index (EWIPI) including SDII (simple daily intensity index), defined as the ratio of annual total precipitation to the number of wet days (≥ 1 mm), representing mean rainfall intensity on wet days. Third, extreme precipitation magnitude indices such as PRCPTOT (annual total wet-day precipitation), RX1DAY (annual maximum 1-day precipitation) and RX5DAY (annual maximum consecutive 5-day precipitation), which quantify the magnitude of extreme precipitation events.

The selection of these eight indices (CDD, CWD, PRCPTOT, RX1DAY, RX5DAY, R10, R30 and SDII) follows the ETCCDI/WMO framework and ensures comparability with previous climate-change studies in the region and globally (Ezaz et al. 2022). Overall, these indices provide a comprehensive description of extremes by capturing the frequency, intensity, duration and magnitude of precipitation events. In line with the ETCCDI classification (Alexander et al. 2006), the indices used here are absolute- and threshold-based; no percentile-based indices were employed, and therefore no separate climatological baseline (e.g., 1971–2000) was required for their computation.

2.4 | Trend Analysis

The Mann–Kendall (MK) test and Sen's slope estimator are non-parametric statistical methods widely used for detecting trends in hydroclimatic time series. Such metrics do not require the data to follow a normal distribution, are relatively

robust to outliers and measurement errors, and provide reliable assessments of long-term trends in extreme weather indices (Dai et al. 2023). In addition to gradual trends, it is important to identify sudden shifts or breakpoints within the time series. Extreme precipitation indices may exhibit abrupt changes in response to large-scale climate variability (e.g., ENSO, and IOD), regional land-use change or phases of accelerated warming. Detecting such shifts improves understanding of when critical transitions occur and strengthens the attribution of observed patterns to specific climatic drivers, and reduces the risk of biased long-term trend estimates that can arise if structural breaks are ignored. This, in turn, provides valuable information for climate adaptation and water-resources management.

In this study, the MK test and Sen's slope estimator were applied to analyse temporal trends in the eight selected EPIs over the period 1980–2017. The MK test yields a Z statistic, where a positive value ($Z>0$) indicates an upward trend and a negative value ($Z<0$) indicates a downward trend. The magnitude of $|Z|$ is used to assess statistical significance: $|Z|\geq 1.96$ indicates significance at the 95% significance level, $|Z|\geq 2.58$ at the 99% level and $|Z|\geq 3.29$ at the 99.9% level (Wang et al. 2022). In addition to the classical Mann–Kendall trend test, the sequential Mann–Kendall (SQ–MK) test and the Pettitt test were applied to detect potential change points (abrupt shifts) in the time series of EPIs. The SQ–MK test is an extension of the MK procedure that uses forward and backward cumulative statistics to identify the timing of possible regime shifts, whereas the Pettitt test is a non-parametric rank-based test for single change points in a series (Wang et al. 2022; Chaulagain et al. 2025).

Given that 160 separate hypothesis tests were performed (8 indices × 20 stations), the Benjamini–Hochberg false discovery rate (FDR) procedure was applied at $\alpha=0.05$ to control for multiple testing (Benjamini and Hochberg 1995). This approach controls the expected proportion of false positives among the rejected null hypotheses and provides a more conservative assessment of significance than unadjusted p-values. FDR-adjusted p-values were obtained using the step-up procedure in which all p-values are ranked in ascending order, each p-value is compared with the critical value $(i/m)\times\alpha$, where i is the rank and m is the total number of tests, and the null hypothesis is rejected for all p-values satisfying $p\leq \mathrm{p_FDR}$.

2.5 | Coherence Analysis

2.5.1 | Correlation Assessment Using Pearson Matrix and Mantel Tests

To examine the linear relationships among the eight chosen extreme precipitation indices, Pearson correlation analysis was employed. This technique measured the strength and significance of pairwise associations between the indices by calculating correlation coefficients. Furthermore, the Mantel test was used to assess the association between large-scale climate drivers (IOD, ENSO, SUNSPOT and AO) and extreme precipitation indices by correlating their respective distance matrices. The significance of the correlations was determined through permutation testing and the findings were illustrated using visualisation tools such as heatmaps and network diagrams.

The Mantel test was chosen because it effectively evaluates the correlation between two distance matrices while accounting for spatial autocorrelation, a common feature in environmental datasets (Borcard and Legendre 2012). In our implementation, Euclidean distance matrices were computed for both the extreme precipitation indices and climate drivers using scipy.spatial.distance.pdist. This function computes pairwise Euclidean distances between observations to create distance matrices for the Mantel test analysis. The test statistics were calculated as the Spearman correlation between these distance matrices. Statistical significance was determined through 1000 permutations, where the null hypothesis assumes no relationship between the matrices. The p-value represents the proportion of permuted correlations that exceeded the observed correlation in absolute value, providing a robust assessment of statistical significance without assumptions about data distribution. All analyses were performed in Python (version 3.12.7) within Jupyter Notebook (version 7.2.2) running on the Anaconda platform (Ren et al. 2024).

2.5.2 | Geo-Detector Method (GDM) and Uncertainty Analysis

The GDM is a powerful statistical approach used to identify and measure spatially stratified heterogeneity and the factors driving it, based on concepts rooted in variance analysis. The fundamental premise is that if an independent variable has a significant effect on a dependent variable, the spatial or temporal distribution patterns of both variables should be similar. GDM consists of four key analytical tools: the Risk Detector, Factor Detector, Ecological Detector, and Interaction Detector (Liu et al. 2021).

The Risk Detector assesses whether significant differences exist in the mean values of the dependent variable across different strata. The Factor Detector quantifies the explanatory power of each independent variable on the dependent variable using q-statistics, where higher q-values indicate stronger influence. The Ecological Detector compares the relative importance of two factors. The Interaction Detector evaluates whether the combined effect of two factors enhances, weakens, or operates independently on the dependent variable (Liu et al. 2021).

In this research, the Factor Detector was employed to assess how major climatic drivers influence various extreme precipitation indices. This process involved calculating the q-statistic, which indicates the proportion of the total variance in the dependent variable that can be explained by an independent variable. A larger q-value reflects a stronger explanatory power. The q-statistic is calculated using the following formula:

$$q = 1 - \frac{\sum_{i=1}^{L} n_i \sigma_i^2}{N\sigma^2} \tag{1}$$

Here, i represents the layers of classification strata for the variable X, n_i represents the number of observations within stratum in layer i, σ_i^2 is the variance of Y within layer i, N refers total

number of observations, while σ^2 denotes the overall variance of Y. In this study, **Y** represents the dependent variable, specifically the extreme precipitation indices (CDD, CWD, R10, R30, SDII, PRCPTOT, RX1DAY and RX5DAY), while **X** represents the independent variable, specifically the climate drivers (IOD, ENSO, SUNSPOT and AO). L represents the total number of layers used for stratification. In this study, $L = 4$ layers based on quantile classification.

The q-statistics range from 0 to 1, where a value of 0 indicates no explanatory power, and 1 indicates complete explanation of spatial variance. Statistical significance of q-values was tested using F-statistics at $p < 0.05$, $p < 0.01$ and $p < 0.001$ levels (denoted by *, ** and ***, respectively). This approach enabled identification of the dominant climatic drivers influencing extreme precipitation patterns across Bangladesh (Liu et al. 2021). By calculating the q-values and assessing their statistical significance, the analysis identified which climatic drivers exert the strongest influence on each category. The findings were visually represented using radar charts.

To quantify uncertainty in Geo-Detector q-statistics arising from the 38-year observational period, bootstrap resampling analysis was performed with 10,000 iterations (Davison and Hinkley 1997). For each iteration, the dataset was resampled with replacement, and q-statistics were recalculated for all 32 EPI-driver combinations (8 EPIs × 4 drivers). The 95% confidence intervals were derived using the percentile method, extracting the 2.5th and 97.5th percentiles of the bootstrap distribution (DiCiccio and Efron 1996). The choice of 10,000 iterations exceeds standard requirements of 1000–2000 iterations to ensure robust confidence interval estimation despite limited temporal extent and the complex spatial stratification characteristic of Geo-Detector analysis (Davison and Hinkley 1997; DiCiccio and Efron 1996). Preliminary convergence testing demonstrated that bootstrap mean q-values varied by less than 0.01 between 500-iteration and 10,000-iteration analyses, confirming computational stability and justifying the conservative iteration count.

2.5.3 | PWC Method

Wavelet Transform Coherence (WTC) is an effective approach for examining how the relationship between two time series evolves over time and across frequency bands (Wu et al. 2022). However, standard bivariate WTC can yield misleading inferences when both series are influenced by additional common drivers. To address this limitation, Hu and Si (2016) proposed the PWC method, which controls for the confounding effects of one or more auxiliary variables and thus provides a clearer assessment of the direct association between two target series. Unlike Multivariate Wavelet Coherence (MWC) (Hu and Si 2016), PWC not only evaluates the significance and persistence of coherence but also resolves phase differences, enabling identification of lead–lag relationships. In this study, PWC was applied to quantify nonlinear, time-varying interactions between extreme precipitation indices and climate drivers, allowing assessment of cyclical behaviour and delayed responses within the climate system.

Sensitivity to the wavelet bandwidth parameter was evaluated using three dj parameter values in the biwavelet R package (Gouhier et al. 2018): $dj = 1/12$ (standard resolution), $dj = 1/6$ (coarser resolution) and $dj = 1/24$ (finer resolution). The dj parameter governs the spacing between discrete scales in the

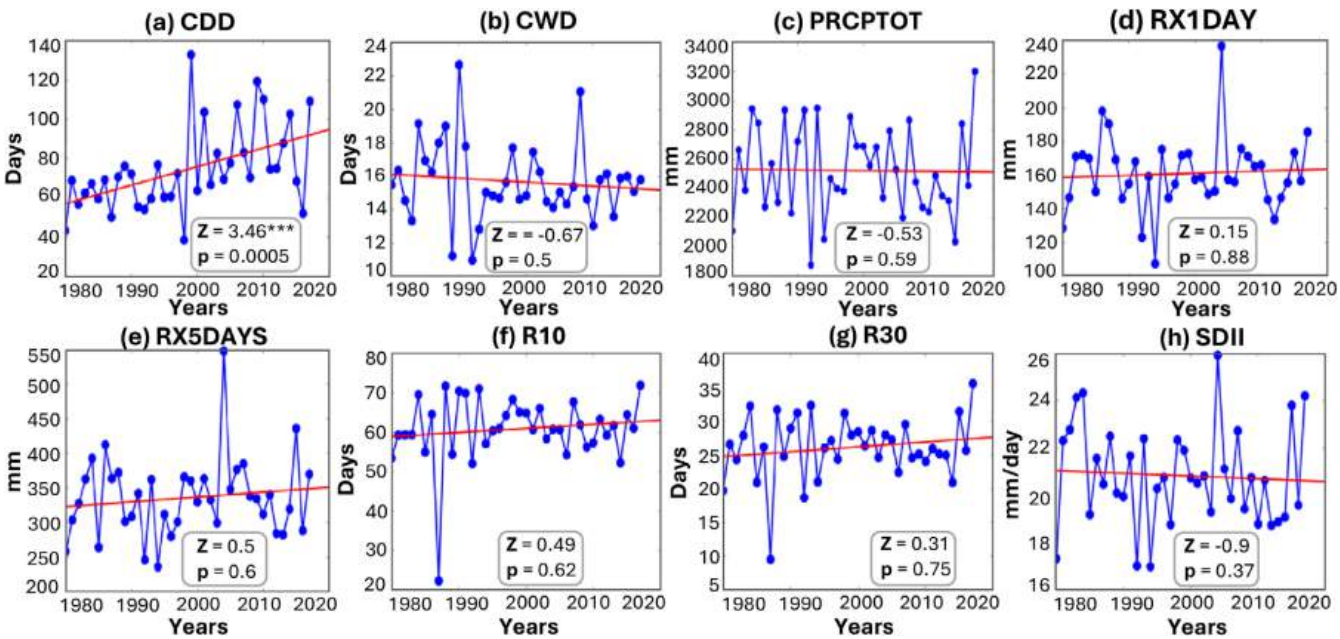

FIGURE 2 | Temporal evolution of eight EPIs in Bangladesh during 1980–2017. The blue line with markers represents annual observed values, while the red line indicates the Sen's slope trend. Subplots report the MK Z statistic and p-value, where positive Z denotes increasing trends and negative Z denotes decreasing trends. Asterisks denote statistical significance (*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$). Values represent regional averages across all available meteorological stations, highlighting the overall national-scale behaviour of precipitation extremes. [Colour figure can be viewed at wileyonlinelibrary.com]

continuous wavelet transform. Robustness was quantified using the coefficient of variation (CV) of the mean significant coherence across the three configurations; CV < 10% was taken to indicate results that are stable and insensitive to *dj* selection.

3 | Results

3.1 | Temporal Trends in EPIs

In the study, the temporal trends of daily rainfall ranges are examined in Bangladesh. The changes in temporal trends of various frequency ranges are demonstrated for 38 years in Figure 2. The annual rainfall data of 20 stations are averaged to obtain the time series of annual average rainfall of Bangladesh for the period 1980–2017. Table 2 summarises the temporal trends of eight EPIs in Bangladesh for 1980–2017.

To clarify the methodology, EPIs were first computed independently for each of the 20 meteorological stations using their respective daily precipitation records for 1980–2017. All EPIs were derived from the quality-controlled and SNHT-verified homogeneous daily precipitation series described in Section 2.2; no raw, non-homogenised data were used in the trend analysis. No averaging of raw precipitation data was performed prior to index computation. After deriving station-level EPI time series,

TABLE 2 | Temporal trends of eight EPIs in Bangladesh.

Index	Units	Regional trend range	Regional average trend	Units	*Z*	*p*
CDD	days	9.1~9.94	9.52	days/decade	**3.46*** **	0.0005
CWD	days	−0.64~0.19	−0.22	days/decade	−0.67	0.5
PRCPTOT	mm	−4.76 to −4.68	−4.72	mm/decade	−0.53	0.59
RX1DAY	days	0.87~1.7	1.28	days/decade	0.15	0.88
RX5DAYS	days	6.68~7.52	7.09	days/decade	0.5	0.6
R10	days	0.62~1.45	1.04	days/decade	0.5	0.62
R30	days	0.32~1.15	0.74	days/decade	0.31	0.75
SDII	mm/day	−0.0004 to −0.0002	0.0003	mm/day/decade	−0.9	0.37

Note: * indicates significance at the 95% confidence level; ** at 99%; and *** at 99.99%. Bold values indicate significance level. All *p*-values are FDR-corrected using Benjamini-Hochberg method ($\alpha = 0.05$) to control for multiple testing across 20 stations × 8 indices.

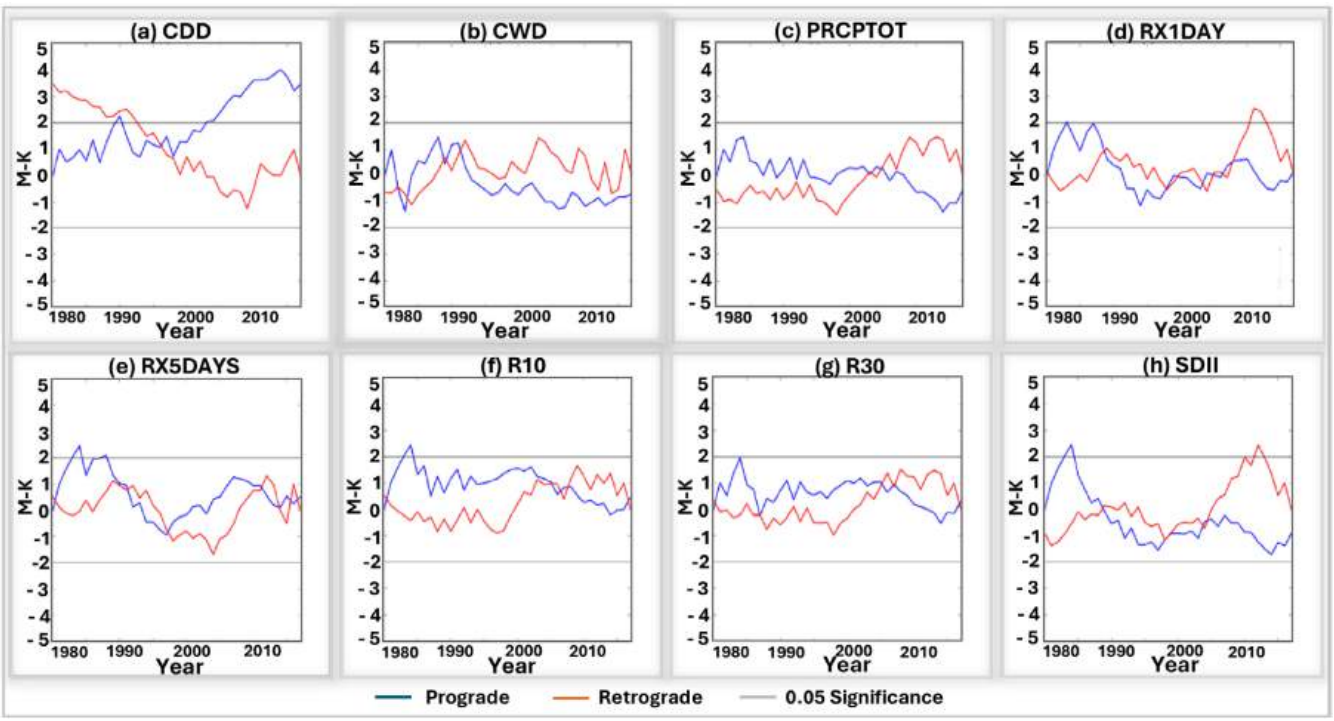

FIGURE 3 | Sequential Mann–Kendall test results for eight EPIs in Bangladesh during 1980–2017. The blue line (Prograde) shows the forward sequential MK statistic, while the red line (Retrograde) shows the backward statistic. The intersection of the two curves denotes the approximate year of a potential structural shift (mutation point). The horizontal black lines mark the 95% significance thresholds (±1.96), when either curve crosses these lines, the corresponding trend is considered statistically significant. [Colour figure can be viewed at wileyonlinelibrary.com]

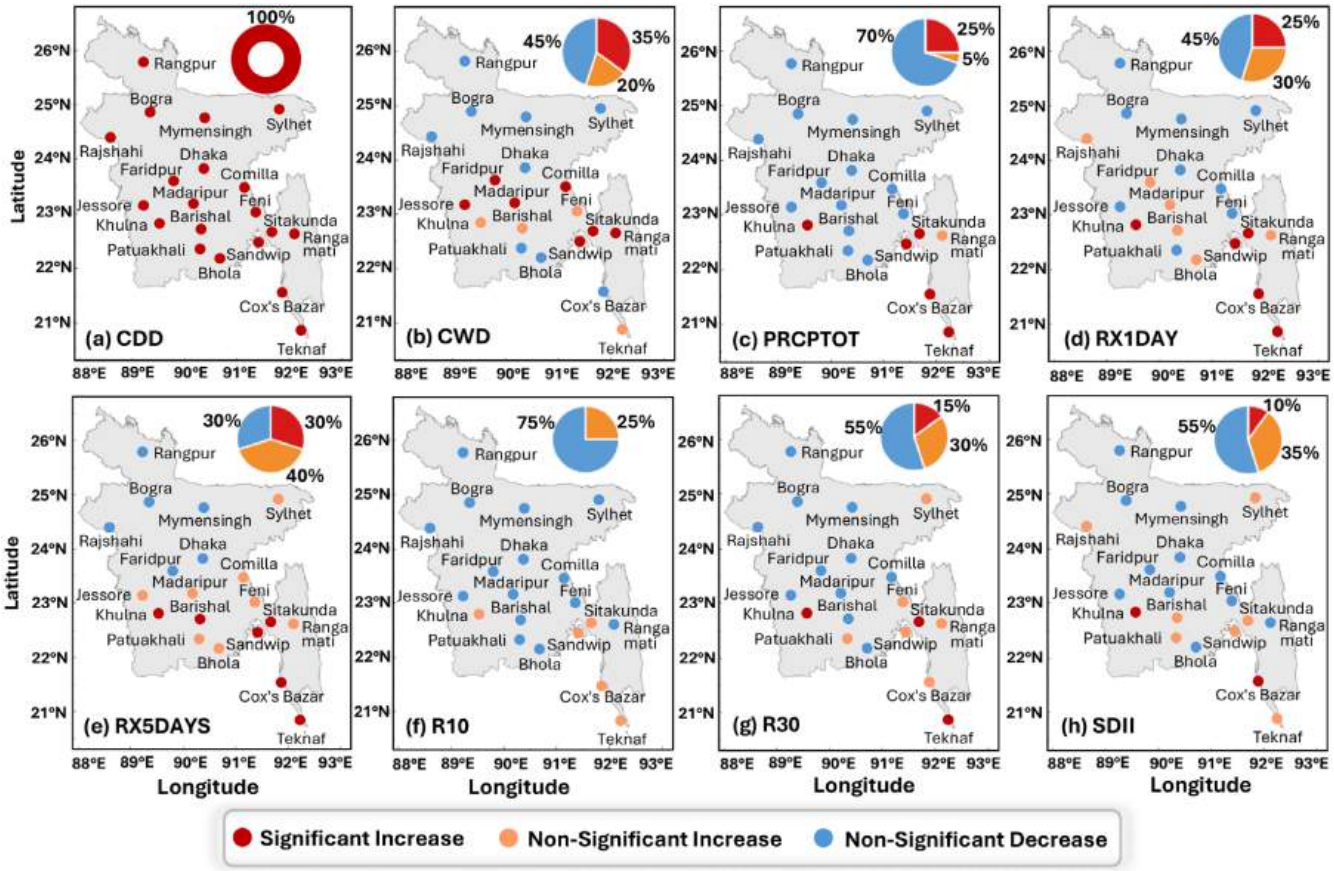

FIGURE 4 | Spatial distribution of station-level trends in eight EPIs across Bangladesh during 1980–2017. Each map shows the location-specific Mann–Kendall test results, classified as significant increase (red circles), non-significant increase (orange circles), significant decrease (blue circles), and non-significant decrease (grey circles). The accompanying pie charts summarise the proportion of stations falling into each trend category for the respective index. These results highlight distinct spatial heterogeneity in precipitation extremes. All significance markers (*) indicate FDR-corrected p-values ($\alpha = 0.05$, Benjamini-Hochberg method) accounting for multiple testing across 160 hypothesis tests. [Colour figure can be viewed at wileyonlinelibrary.com]

national-scale temporal patterns shown in Figure 2 were obtained by averaging the already computed station-level indices. This approach preserves local extremes, avoids dilution effects that occur when raw precipitation is averaged across stations, and follows recommended ETCCDI practice for analysing precipitation extremes. Therefore, the plots in Figure 2 represent aggregated EPI values, not averaged rainfall data.

The most robust result is the significant increase in CDD, which rose by an average of 9.52 days per decade ($Z = 3.46$, $p < 0.001$). This indicates intensifying dry spells and is consistent with an elevated risk of drought. In contrast, CWD exhibited a slight, non-significant decrease of −0.22 days per decade, suggesting a modest reduction in the persistence of wet periods. The PRCPTOT showed a weak but negative trend (−4.72 mm per decade, $Z = -0.53$, $p = 0.59$), and the SDII also displayed a non-significant decline, together pointing towards a weakening of the mean monsoon rainfall regime.

Short-duration extremes, represented by RX1DAY and RX5DAY, increased slightly (1.28 and 7.09 mm per decade, respectively), but neither trend was statistically significant. Similarly, heavy and very heavy precipitation days (R10 and R30) showed weak upward tendencies, with increases of 1.04 and 0.74 days per decade, respectively, again without reaching statistical significance. Overall, these results suggest an emerging 'dual' rainfall regime: on the one hand, significantly longer dry periods (increasing CDD) combined with declining mean precipitation (PRCPTOT and SDII), and on the other hand, modest, statistically non-significant indications of more frequent or intense short-duration heavy rainfall events (RX1DAY, RX5DAY and R30). This pattern implies that Bangladesh's precipitation climate is becoming more irregular, characterised by extended drought-like spells punctuated by occasional intense rainfall episodes.

In the sequential Mann–Kendall (SQ–MK) plots (Figure 3), the two horizontal black lines denote the 95% confidence thresholds (±1.96); intersections of the SQ–MK curves with these lines indicate statistically significant trends. For CDD, the forward sequence crosses the upper significance boundary in the late 1990s, consistent with the Pettitt test, which identifies a change point in 1998. This confirms a significant upward trend in CDD and indicates an intensification of dry spells across Bangladesh.

For CWD, the Pettitt test suggests a shift in 1991; however, both the forward and backward SQ–MK curves remain within

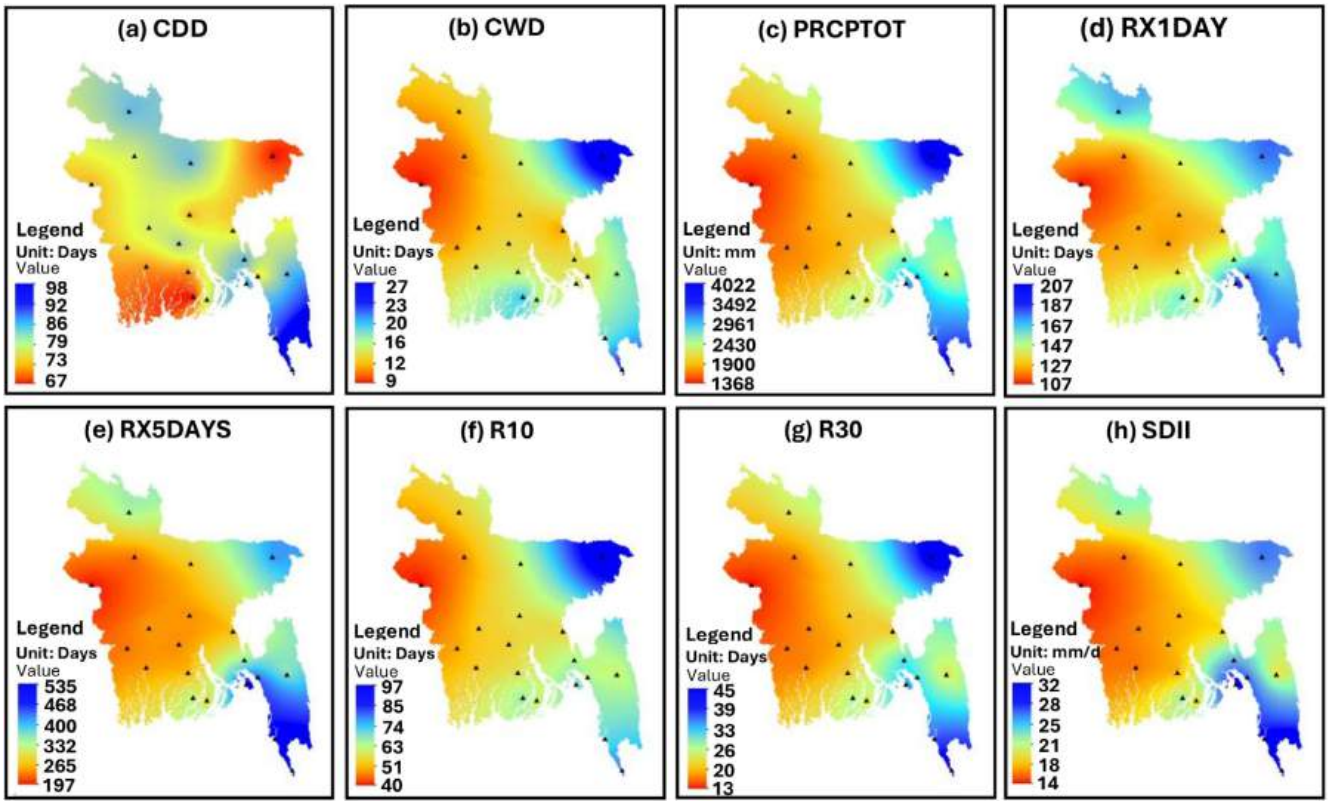

FIGURE 5 | Spatial distribution of the long-term mean annual values (1980–2017) of eight EPIs across Bangladesh. Station-level means were interpolated using Inverse Distance Weighting (IDW) to produce gridded raster surfaces, highlighting distinct regional contrasts, with dry extremes (CDD) strongest in the southeast and wet extremes (CWD, PRCPTOT, RX1DAY, RX5DAY, R10, R30 and SDII) concentrated in the northeast and southeast regions. [Colour figure can be viewed at wileyonlinelibrary.com]

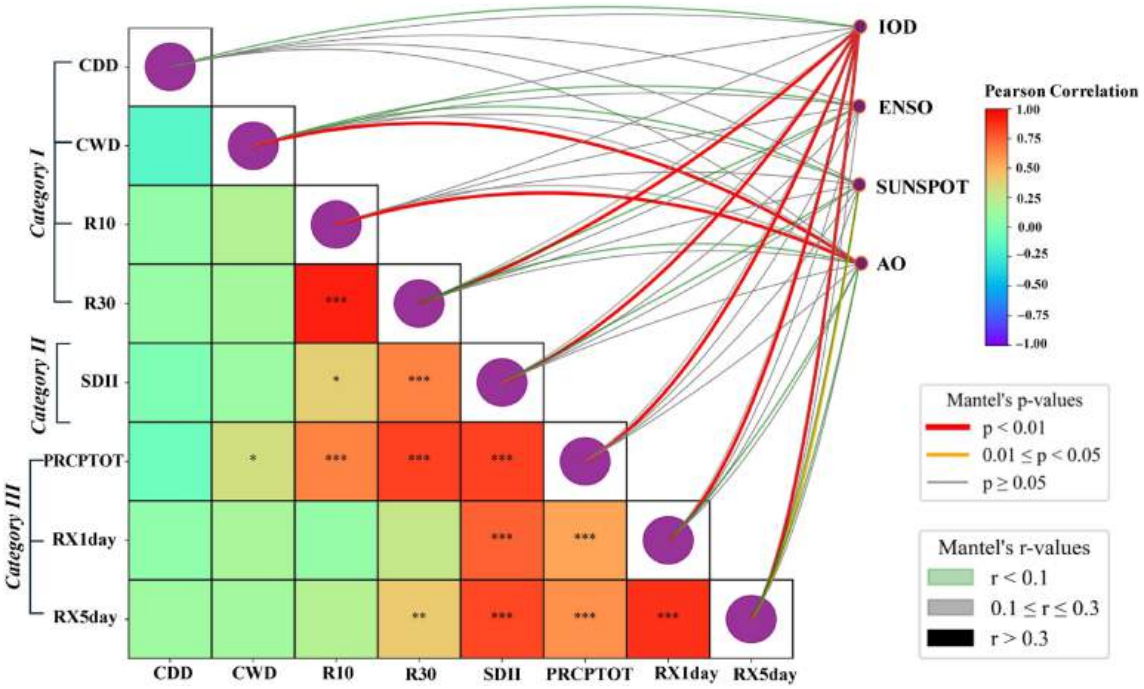

FIGURE 6 | Pearson correlations among eight extreme precipitation indices (left panel) and Mantel test associations with large-scale climate drivers (right panel). Colours represent correlation strength from −1 to +1 based on the colour bar, with asterisks denoting significance levels (*$p \leq 0.05$, **$p \leq 0.01$, ***$p \leq 0.001$). Curved connections indicate Mantel test results: line colour corresponds to p-value (red = $p < 0.01$, orange = 0.01–0.05, grey = not significant), while line opacity indicates Mantel's r strength. Purple circles along the diagonal represent the indices themselves. [Colour figure can be viewed at wileyonlinelibrary.com]

the confidence bounds, implying that this change is not statistically robust. A similar pattern is observed for PRCPTOT: although the Pettitt test detects a change point in 2005, the SQ–MK curves do not cross the significance thresholds, indicating that the apparent decline in total rainfall is not statistically significant.

Among the extreme rainfall indices, RX1DAY and SDII exhibit Pettitt-detected change points around 1988, RX5DAY around 1997, and R10 and R30 around 1987. In all these cases, the SQ–MK curves fluctuate near, but do not consistently exceed, the critical limits, suggesting that these shifts are more indicative of short-term variability than of persistent, statistically significant long-term changes. Overall, the combined evidence from the SQ–MK and Pettitt tests shows that only CDD exhibits a statistically significant monotonic trend. The remaining indices display temporal shifts without strong statistical support, underscoring the prominence of drying tendencies relative to coherent wetting signals in Bangladesh's precipitation regime.

3.2 | Spatial Patterns and Trends of EPIs

Figure 4 illustrates the spatial distribution of station-level trends in the eight extreme precipitation indices across Bangladesh. The results reveal a pronounced dominance of increasing CDD. After application of the FDR correction (Table S3), 95% of stations (19/20) still exhibit significant increasing trends, compared with 100% under uncorrected *p*-values, confirming the widespread lengthening of dry spells across the country. In contrast, CWD displays a heterogeneous spatial response: 45% of stations show significant increases, 35% exhibit non-significant decreases, and 20% show non-significant increases. This pattern reflects spatially variable

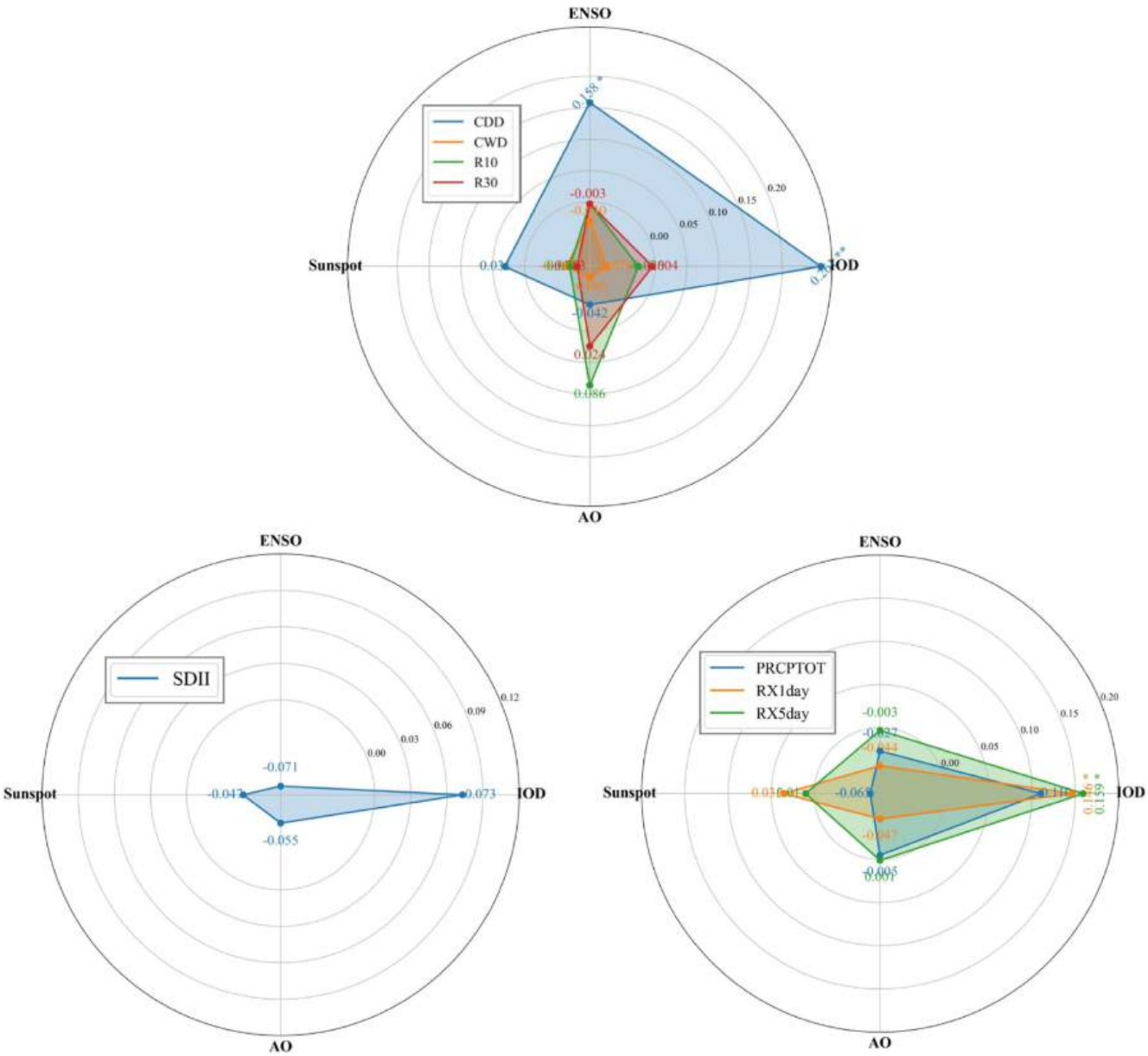

FIGURE 7 | Relative influence of different climatic drivers on selected EPIs. Asterisks *, ** and *** indicate statistical significance at the 0.05, 0.01 and 0.001 levels, respectively. [Colour figure can be viewed at wileyonlinelibrary.com]

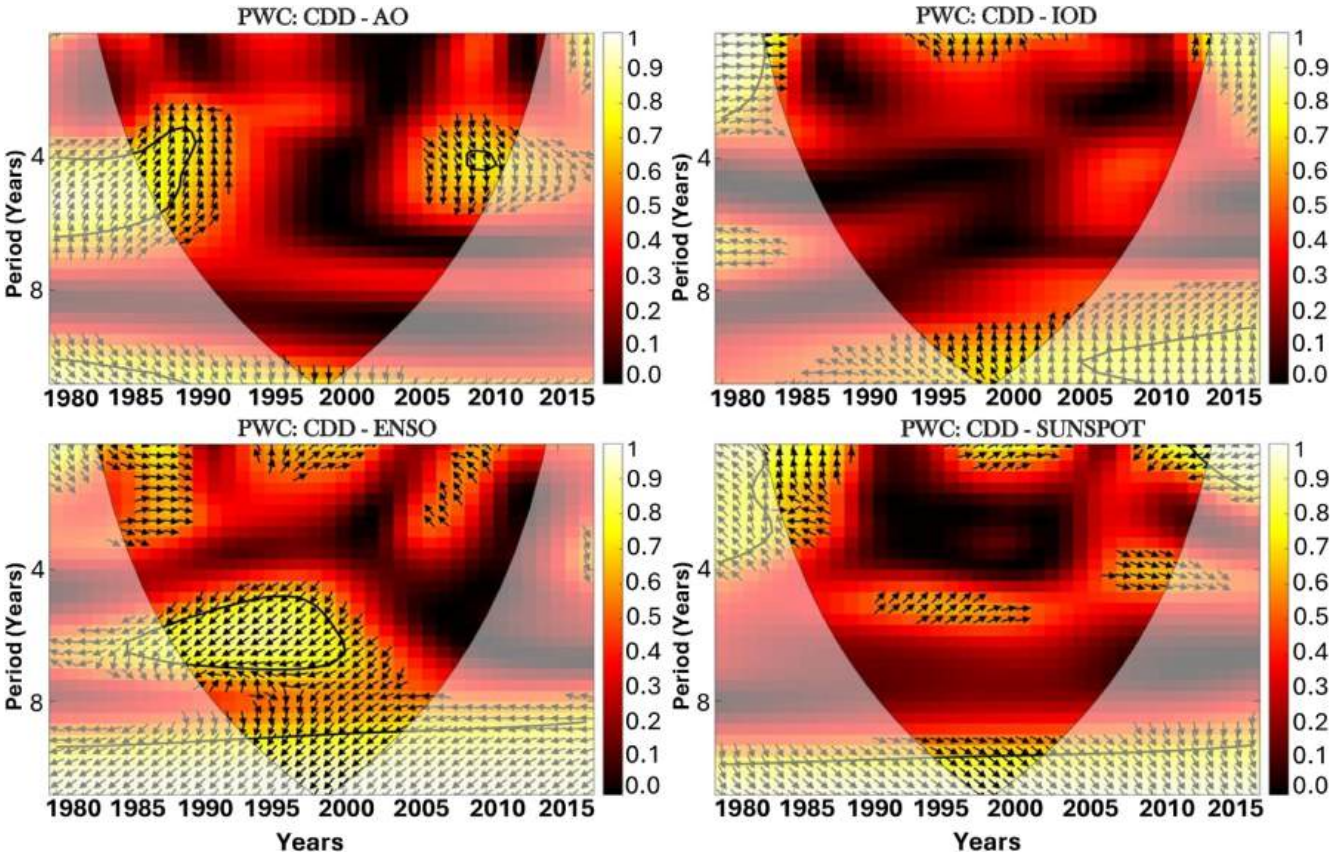

FIGURE 8 | Partial wavelet coherence (PWC) analysis between CDD and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

changes in wet-spell duration, with localised increases in the northeast and reductions elsewhere. PRCPTOT shows a general weakening of annual wet-day precipitation, with 70% of stations exhibiting significant decreases and only 5% showing increases, consistent with declining seasonal rainfall in many regions. SDII similarly indicates widespread reductions in mean rainfall intensity, with 55% of stations showing significant decreases.

The heavy-rainfall indices exhibit notable spatial variability. RX1DAY trends are divided, with 45% of stations showing significant increases and 30% showing significant decreases. RX5DAY displays a similarly mixed pattern, with equal proportions of stations (30%) indicating significant increases and decreases. These mixed signals underscore the localised nature of multi-day extreme rainfall events. For event-frequency indices, R10 is dominated by decreasing trends: 75% of stations show significant declines. In contrast, R30 retains significant increasing trends at only 15% of stations (3/20) after FDR correction, compared with 45% (9/20) prior to correction.

Figure 5 illustrates the spatial distribution of the long-term annual mean values for the EPIs across Bangladesh. The results of the interpolation reveal distinct regional contrasts governed by the country's unique topographical features and the subtropical monsoon system. For the CDD, the highest values are concentrated in the southeastern coastal belt and the western regions. The absolute peak of dry spell duration occurs in Teknaf (98 days) and Cox's Bazar (82 days), reflecting a prolonged and intense distinct dry season in the southeast. In contrast, the lowest CDD values are observed in Patuakhali (67 days) and Sylhet (~67 days), where rainfall is more frequent throughout the year.

The CWD follows a different spatial logic, peaking sharply in the Northeast. Sylhet records the highest mean of approximately 28 days, while the lowest values occur in Rajshahi (9 days), confirming the strong orographic and monsoonal influence on the northeastern hills. The PRCPTOT follows this gradient, exceeding 4000 mm in Sylhet but dropping significantly in the northwest, with Rajshahi receiving only 1371 mm annually.

Similar spatial gradients are observed for intensity indices such as RX1DAY and RX5DAY. The heaviest one-day rainfall is concentrated along the coast and the northeast, with Sandwip (207 mm) and Cox's Bazar (536 mm for RX5DAY) exhibiting the highest extremes. Rajshahi consistently displays the lowest values for both indices (108 and 197 mm, respectively). The frequency indices R10 and R30 also peak in Sylhet (98 and 46 days), decreasing to their minimums in Rajshahi (40 and 14 days). Finally, the Simple Daily Intensity Index (SDII) is strongest along the southeast coast, reaching 32 mm/day in Cox's Bazar compared to only 14 mm/day in Rajshahi.

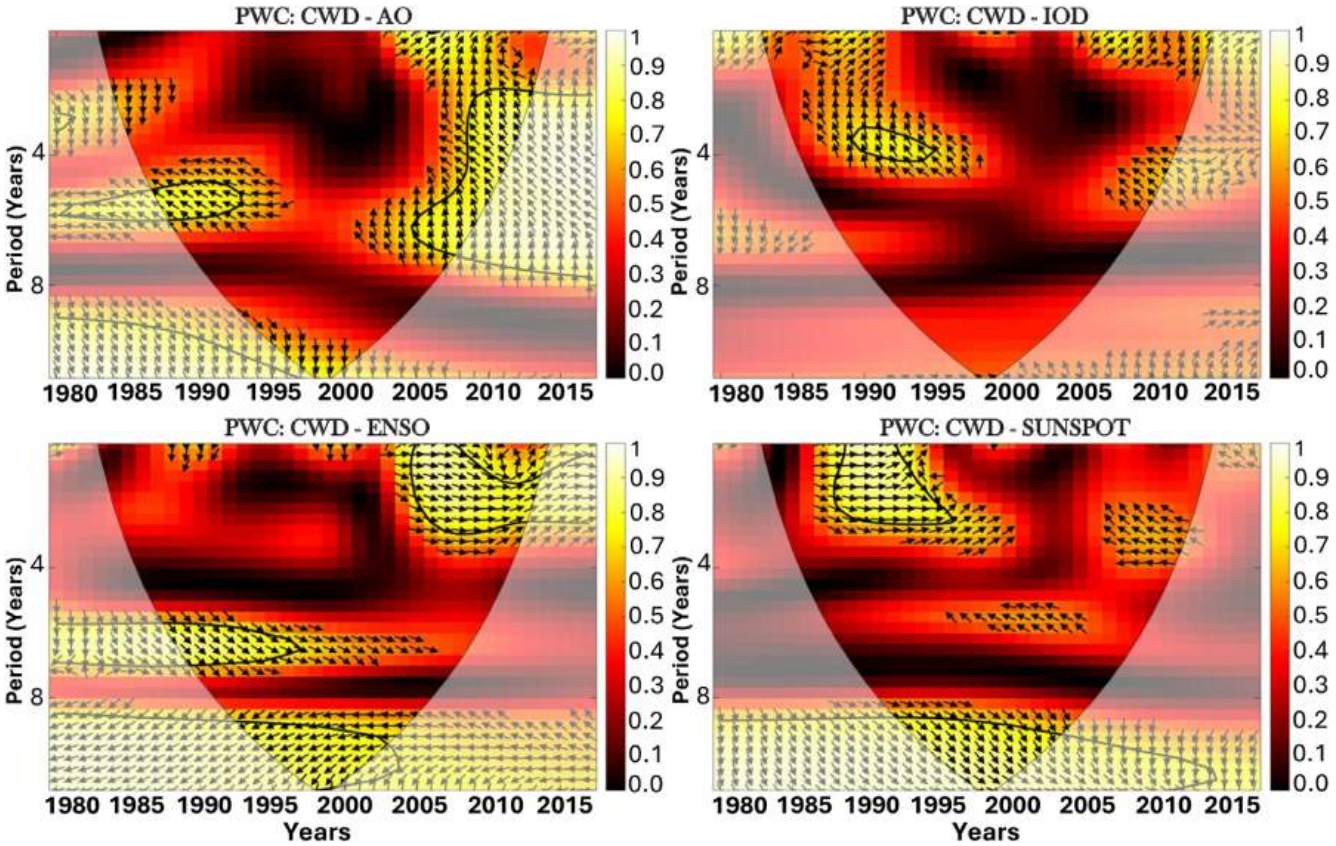

FIGURE 9 | Partial wavelet coherence (PWC) analysis between CWD and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

Overall, these Kriging results highlight a pronounced Northwest–Southeast intensity gradient. While the Northeast and Southeast experience the heaviest and most frequent rainfall, the Southeast is unique in experiencing both extreme wet periods and the longest consecutive dry spells (CDD). Meanwhile, the Northwest remains the most consistently arid region, characterised by lower total rainfall and fewer heavy-precipitation events.

3.3 | Correlation of EPIs With Climatic Drivers

Figure 6 illustrates the interrelationships among the eight EPIs and their associations with large-scale climatic drivers. Clear clustering is evident among the indices: R10, R30, PRCPTOT, RX1DAY, RX5DAY and SDII (Categories II and III) are strongly and positively correlated ($r > 0.6$, $p < 0.001$), underscoring their shared dependence on total rainfall amount and rainfall intensity. By contrast, CDD (Category I) shows significant negative correlations with wetness-related indices such as PRCPTOT, RX1DAY and RX5DAY, reflecting the intrinsic trade-off between prolonged dry spells and extreme rainfall events. CWD exhibits weaker and more mixed correlations, consistent with its localised spatial variability.

The Mantel test indicates strong and statistically significant linkages between EPIs and several large-scale climate modes. ENSO and IOD exert the most consistent influence, displaying robust positive associations with CDD and negative associations with PRCPTOT, RX1DAY and RX5DAY. This pattern suggests that warm-phase ENSO/IOD conditions are often associated with longer dry periods and suppressed rainfall extremes over Bangladesh. ENSO and IOD show moderate but significant associations with PRCPTOT and RX5DAY, whereas AO and SUNSPOT exhibit weaker and largely non-significant associations. Overall, these results indicate that hydroclimatic extremes in Bangladesh are not only internally coherent across indices but are also strongly modulated by large-scale ocean–atmosphere variability, with ENSO and IOD emerging as the dominant drivers of rainfall variability and extremes.

3.4 | Influence of Climatic Drivers on EPIs

Figure 7 presents radar plots summarising the associations between extreme precipitation indices (EPIs) and four large-scale climatic drivers (IOD, ENSO, AO, SUNSPOT). Uncertainty in all q-statistics was quantified using a bootstrap procedure with 10,000 iterations (Table S4). The strongest relationships, expressed with 95% confidence intervals, are: CDD–IOD ($q = 0.315$ [0.112–0.579]), R30–IOD ($q = 0.289$ [0.060–0.570]), and SDII–IOD ($q = 0.288$ [0.053–0.599]). In all cases, the

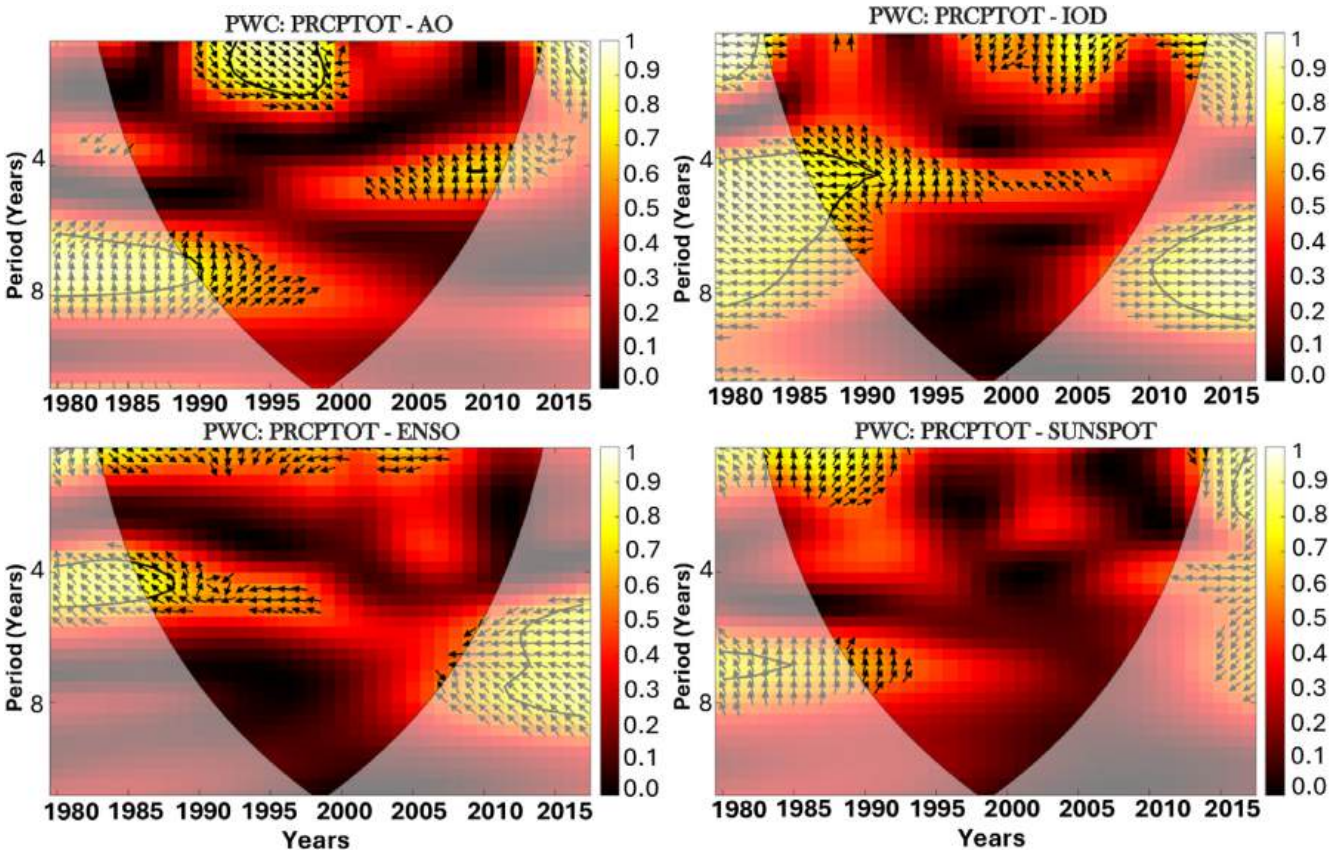

FIGURE 10 | Partial wavelet coherence (PWC) analysis between PRCPTOT and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO, and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: Rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

confidence intervals exclude zero, indicating statistically robust associations despite the relatively short (38-year) record.

For Category I indices (CDD, CWD, R10 and R30), the IOD and ENSO emerge as the dominant modulators. CDD shows positive correlations with both IOD ($r \approx 0.22$) and ENSO ($r \approx 0.08$), suggesting that warm-phase conditions are associated with prolonged dry spells. By contrast, wet-day frequency indices such as R10 and R30 exhibit weaker or negative correlations with IOD and ENSO, consistent with reduced monsoon rainfall during El NIÑO years.

For the Category II index (SDII), correlations with the climate drivers are generally weak, although small negative associations with IOD ($r \approx -0.07$) and ENSO ($r \approx -0.05$) suggest a tendency for mean rainfall intensity on wet days to decrease under warm-phase variability. Category III indices (PRCPTOT, RX1DAY, and RX5DAY) display clearer relationships with IOD and ENSO than with other modes. RX5DAY, for example, is positively correlated with IOD ($r \approx 0.12$) and ENSO ($r \approx 0.07$), whereas PRCPTOT shows a weak but consistent negative correlation with ENSO ($r \approx -0.05$). AO and SUNSPOT exhibit negligible associations with most indices. Overall, the radar plots highlight the dominant role of ENSO and IOD in modulating rainfall extremes in Bangladesh, particularly in regulating dry spells (CDD) and multi-day precipitation events (RX5DAY), while other climatic drivers play weaker and more spatially limited roles.

3.5 | Phase Relationships and Cyclical Patterns Between EPIs and Climatic Drivers

PWC was computed using annual EPI time series derived independently for each station. After the indices (CDD, CWD, RX1DAY, RX5DAY, R10, R30, PRCPTOT and SDII) were calculated at station level following ETCCDI definitions, PWC was applied separately for each station against the corresponding climate-driver series (ENSO, IOD, AO and solar activity). This station-based approach preserves local characteristics of extremes and avoids the smoothing inherent in coherence analyses based on spatially averaged precipitation. Significant coherence regions were then examined across stations to identify spatially consistent teleconnection patterns.

All daily precipitation series underwent quality control and gap-filling (with $<5\%$ missing values) before indices were computed, ensuring that each station contributed a continuous, evenly spaced annual EPI series suitable for wavelet and PWC analysis.

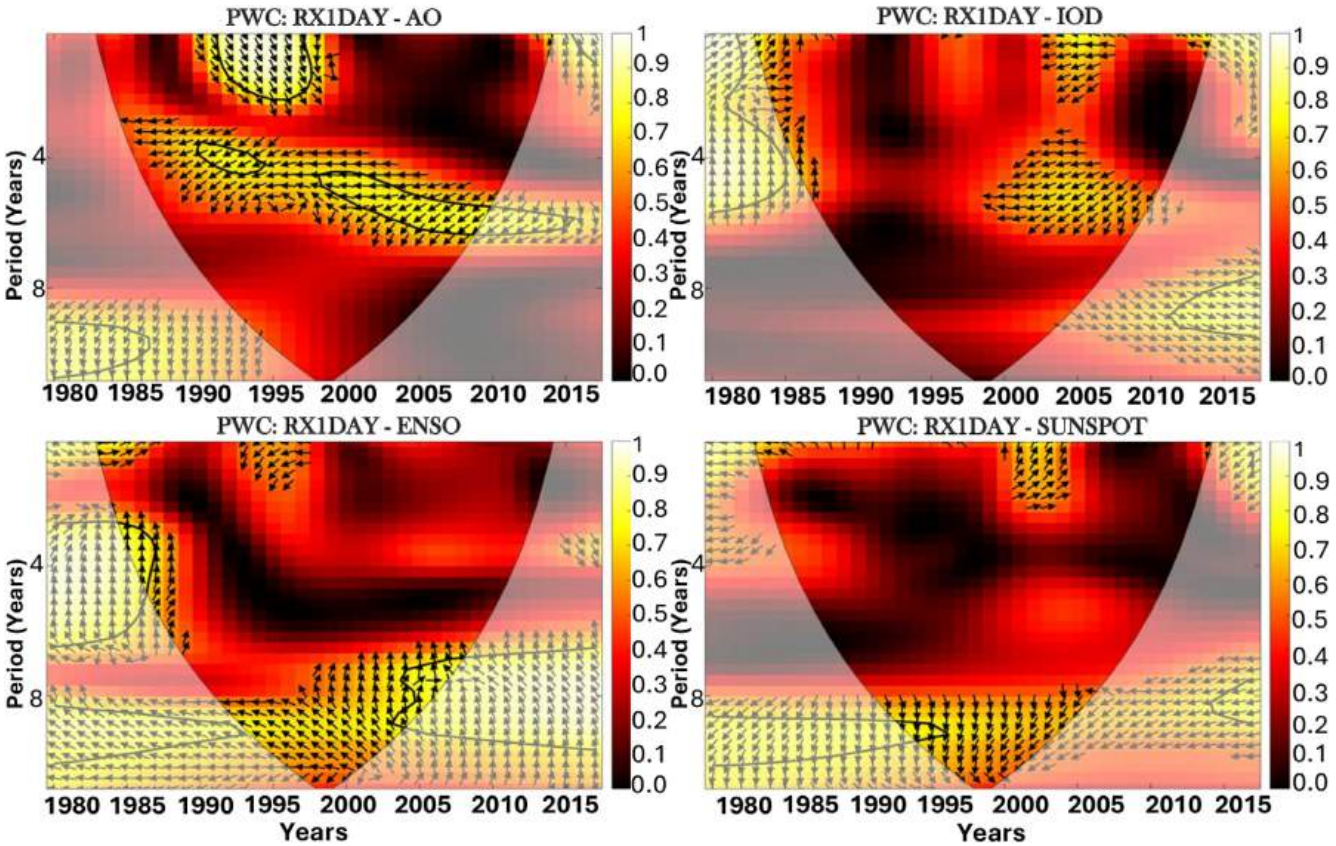

FIGURE 11 | Partial wavelet coherence (PWC) analysis between RX1DAY and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: Rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

The periodicity range of 2–16 years used in Figures 8–15 was chosen to focus on dominant interannual-to-decadal modes relevant to Bangladesh's hydroclimate, particularly ENSO and IOD (≈2–7 years) and the shorter solar-cycle–related variability (~10–12 years), which can be robustly resolved within the 38-year observational record. Lower-frequency oscillations (> 16 years) were excluded because the available data span contains too few full cycles for reliable coherence estimation. Additional y-axis tick marks were included to improve readability of the period axis. Figures 8–15 show PWC results for three groups of EPIs: (1) PFCIs—CDD, CWD, R10 and R30; (2) EWIPI—SDII; and (3) magnitude-based EPIs—PRCPTOT, RX1DAY and RX5DAY, each analysed against AO, ENSO, IOD and SUNSPOT indices. Figure 8 illustrates the PWC between CDD and the selected large-scale climate drivers. Coherence patterns vary across time and periodicity bands. ENSO exhibits the most consistent and statistically significant coherence with CDD, particularly within the 2–8-year band during the late 1980s to early 2000s, where significant regions are enclosed by confidence contours and fall largely within the cone of influence. IOD shows intermittent coherence within similar interannual bands, though these regions are less spatially continuous and more temporally fragmented compared to ENSO. In contrast, AO and SUNSPOT display scattered and relatively localised coherence patches without persistent multi-year continuity. Many of these features occur near the cone-of-influence boundary, suggesting cautious interpretation. Overall, the results indicate that interannual variability in CDD is most coherently associated with ENSO-related variability, while the remaining drivers exhibit weaker and more episodic relationships within the resolved 2–16-year band.

The PWC results for CWD indicate that interannual variability is strongly influenced by ENSO and IOD, with significant coherence at 2–8-year periods during roughly 1990–2005 (Figure 9). This suggests that warm-phase ENSO and positive IOD events were associated with enhanced persistence of wet spells over Bangladesh during those years. Relationships with AO and SUNSPOT activity are relatively weak and intermittent, lacking consistent multi-decadal signals. Overall, these patterns indicate that CWD variability at interannual scales is primarily governed by ENSO and IOD.

For PRCPTOT, PWC reveals significant influence from multiple climate drivers across both interannual and decadal bands. ENSO and IOD show strong coherence with PRCPTOT in the 2–8-year range, particularly between 1990 and 2010, indicating that their positive phases are linked to episodes of enhanced rainfall over Bangladesh. AO and SUNSPOT activity display weaker, episodic coherence at shorter periods (2–4 years), implying only transient effects on rainfall. Taken together, these results suggest that ENSO and IOD dominate short-term variability in PRCPTOT.

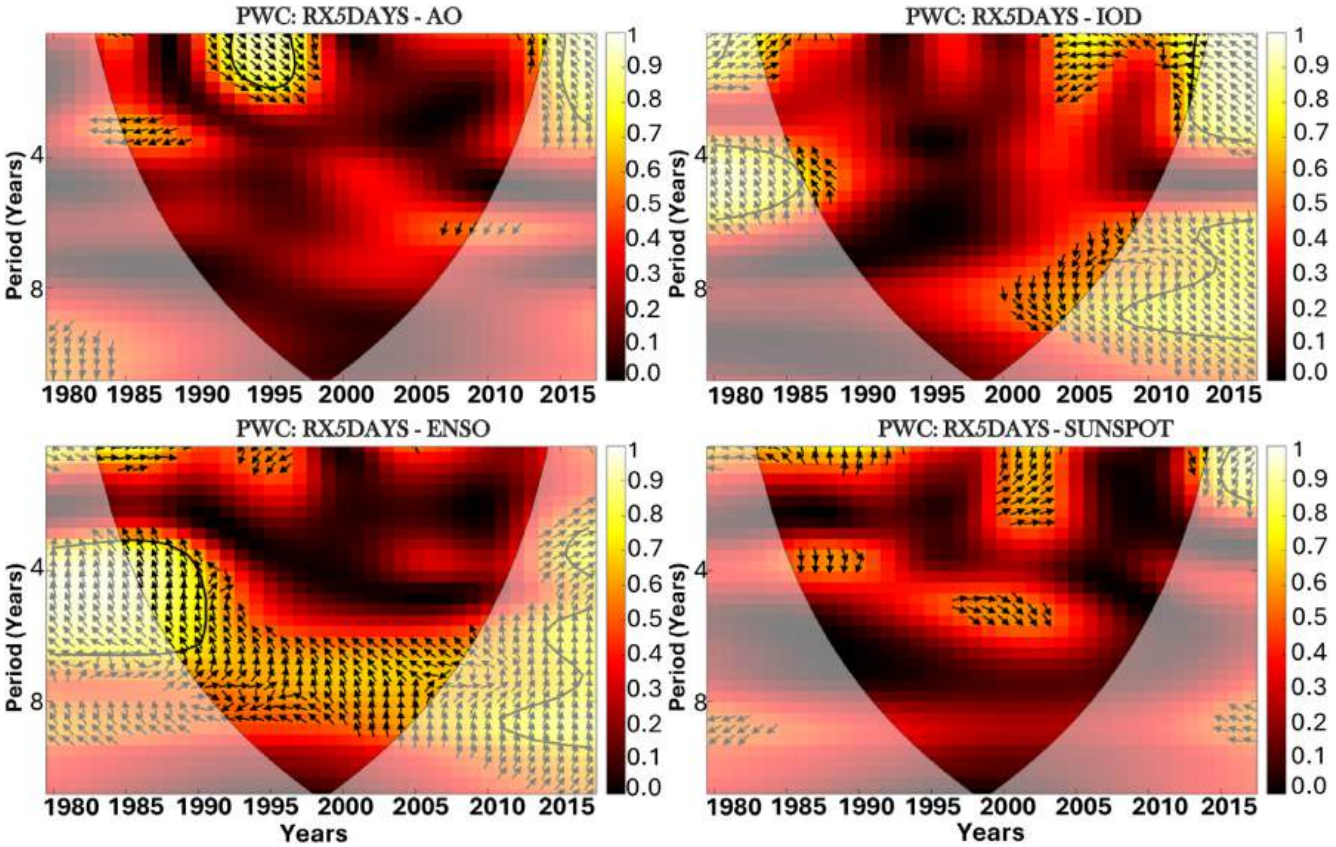

FIGURE 12 | Partial wavelet coherence (PWC) analysis between RX5DAY and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

The relationship between the AO exhibits very strong coherence with RX1DAY, with values approaching 1.0 in the 2–4-year band, peaking near 1995. ENSO shows pronounced coherence with RX1DAY during 1985–1990 and 2000–2010, with coherence values around 0.8, indicating a substantial modulating influence. The IOD displays moderate coherence (≈0.6–0.7) during 2000–2010 at 4–8-year periods. By comparison, SUNSPOT activity is associated with only weak coherence (≈0.6), mainly between 1990 and 2005.

RX5DAY shows a marked association with AO, with coherence values of about 0.9 at 2–4-year periods, especially between 1990 and 2000, underscoring the potential role of AO in modulating multi-day extreme rainfall. RX5DAY demonstrates significant coherence with ENSO at 4–8-year periods, with values around 0.7–0.8, most clearly expressed from the mid-1980s to the late 1990s. Coherence with IOD is weaker (≈0.5–0.6). In contrast, coherence between RX5DAY and solar activity remains low (≈0.5–0.6), implying that SUNSPOT variability exerts little influence on RX5DAY behaviour over Bangladesh.

Coherence between R10 and AO is also notable in the 2–4-year band from approximately 1985 to 2010, with phase directions implying that AO exerts a sustained influence on R10 variability. By contrast, R10–ENSO coherence remains weak throughout most of the record, with only small, isolated patches in the early 1980s and late 2000s, indicating an intermittent and generally limited ENSO impact on R10. The R10–IOD relationship is similarly weak and spatially fragmented, with only sparse coherence patches, pointing to a less systematic role for IOD in modulating R10. Coherence between R10 and SUNSPOT is strongest in the early 1980s, but the phase structure is variable, indicating a complex and likely nonlinear solar influence on R10.

Coherence with AO is significant from about 1995 to 2000 in the 2–4-year band (≈0.8–0.9), implying a consistent AO influence on R30 at interannual scales. In contrast, coherence with ENSO is patchy and moderate in the 2–4-year band, with arrows often pointing leftward and downward, consistent with an ENSO leading, anti-phase relationship, indicating that events may be associated with reduced R30 occurrences at certain times. High coherence with IOD is evident in the 2–4-year band from roughly 1995 to 2010, with phase arrows mainly pointing rightward and upward, indicating a positive influence of IOD, particularly its positive phases, on intense rainfall events. Coherence with solar activity is strongest during 1985–1995 in the 2–4-year band; arrows generally point rightward and upward, indicating a positive, in-phase to slightly SUNSPOT leading relationship, which may reflect a modest enhancing effect of solar variability on very heavy rainfall during these cycles.

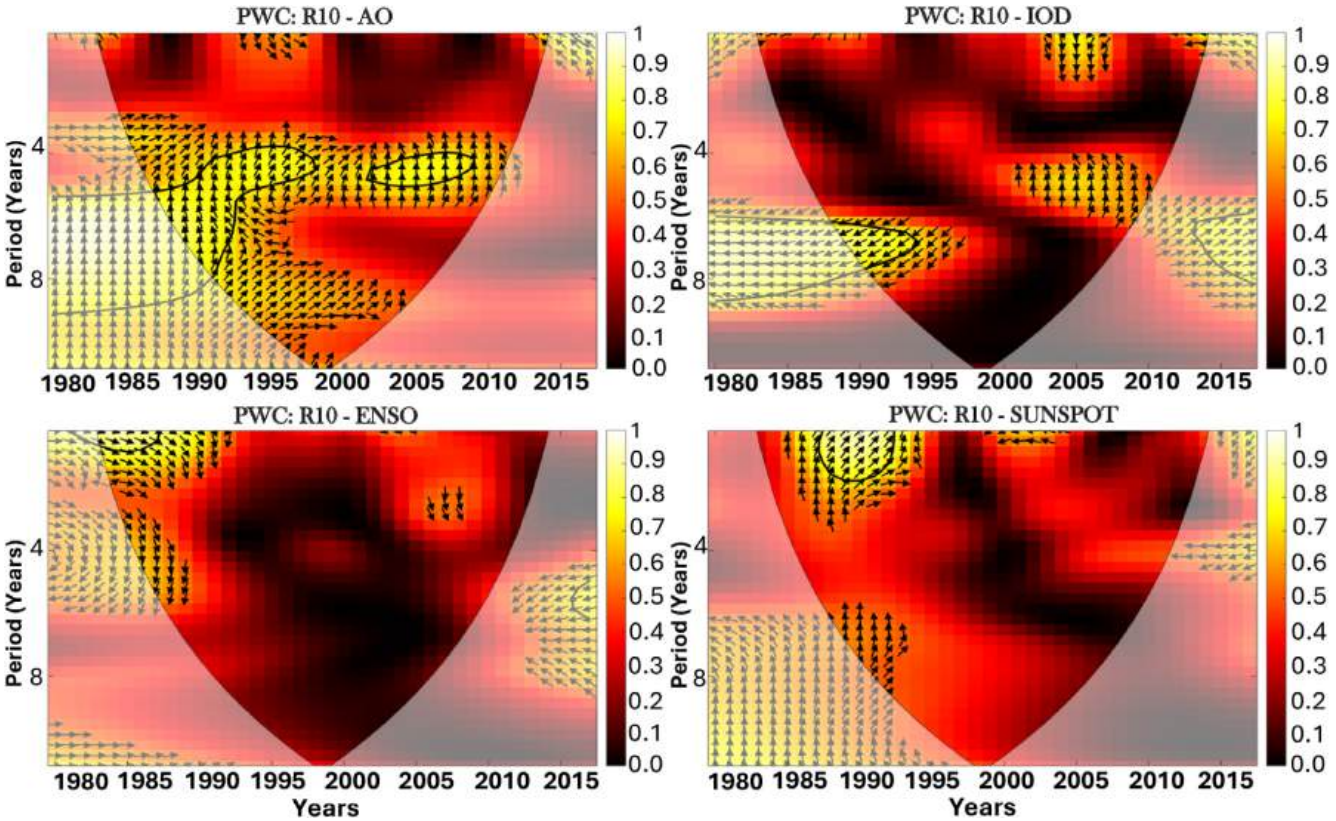

FIGURE 13 | Partial wavelet coherence (PWC) analysis between R10 and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: Rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

SDII shows significant coherence with AO from about 1990 to 2000, with values around 0.8–0.9, suggesting a strong AO influence on daily rainfall intensity over this period. ENSO exhibits moderate coherence with SDII (≈0.6–0.7), implying that ENSO phases affect variability in daily rainfall intensity. IOD demonstrates a moderate to strong relationship with SDII, with coherence values of about 0.7–0.8 in the 2–4-year band. By contrast, solar activity displays negligible coherence with SDII, indicating little to no discernible influence of SUNSPOT variability on daily precipitation intensity over Bangladesh.

4 | Discussion

4.1 | Influence of Regional Geography on Extreme Precipitation

Regional geography strongly conditions extreme precipitation in Bangladesh. Positioned within the South Asian monsoon belt, the country reflects both large-scale ocean–atmosphere controls and sharp physiographic contrasts (Huq et al. 2022). We find a clear meridional gradient: latitude is negatively correlated with CDD ($r = -0.56$, $p < 0.05$), implying shorter dry spells in the southern coastal zone (e.g., Barisal, Khulna) due to greater Bay of Bengal moisture influence, and longer dry spells in the north (e.g., Rajshahi, Rangpur), consistent with elevated drought risk in north-western Bangladesh (Shahid and Osman 2009). Longitude shows only weak associations with EPIs, which is expected given Bangladesh's limited east–west extent; however, urban expansion and floodplain alteration may locally intensify extremes in cities such as Dhaka and Chattogram by reducing storage and buffering capacity (Pandey and Ghosh 2023).

Topography further modulates extremes. Mid-elevation areas (10–30 m) show lower rainfall totals and intensities, with strong negative correlations for PRCPTOT ($r = -0.73$, $p < 0.05$) and SDII ($r = -0.78$, $p < 0.05$), whereas higher-elevation regions (e.g., Chittagong Hill Tracts) exhibit enhanced precipitation linked to orographic uplift and convective development (Shahid and Khairulmaini 2009). At the synoptic scale, weakening or eastward displacement of south-westerlies is typically associated with reduced rainfall over Bangladesh, while the country remains exposed to recurrent flooding and episodic drought, particularly in the northwest (Shahid and Osman 2009). These circulation effects are increasingly modulated by warming-driven shifts in precipitation regimes and intensification of hydroclimatic hazards (Wang et al. 2022).

For elevation, stations were grouped into three altitude bands—0–10 m (coastal/tidal floodplain), 10–30 m (central floodplain), and > 30 m (northeastern highlands and Chittagong Hill Tracts). Table 3 correlations represent within-class relationships rather than a continuous elevation gradient (Das 2021). The

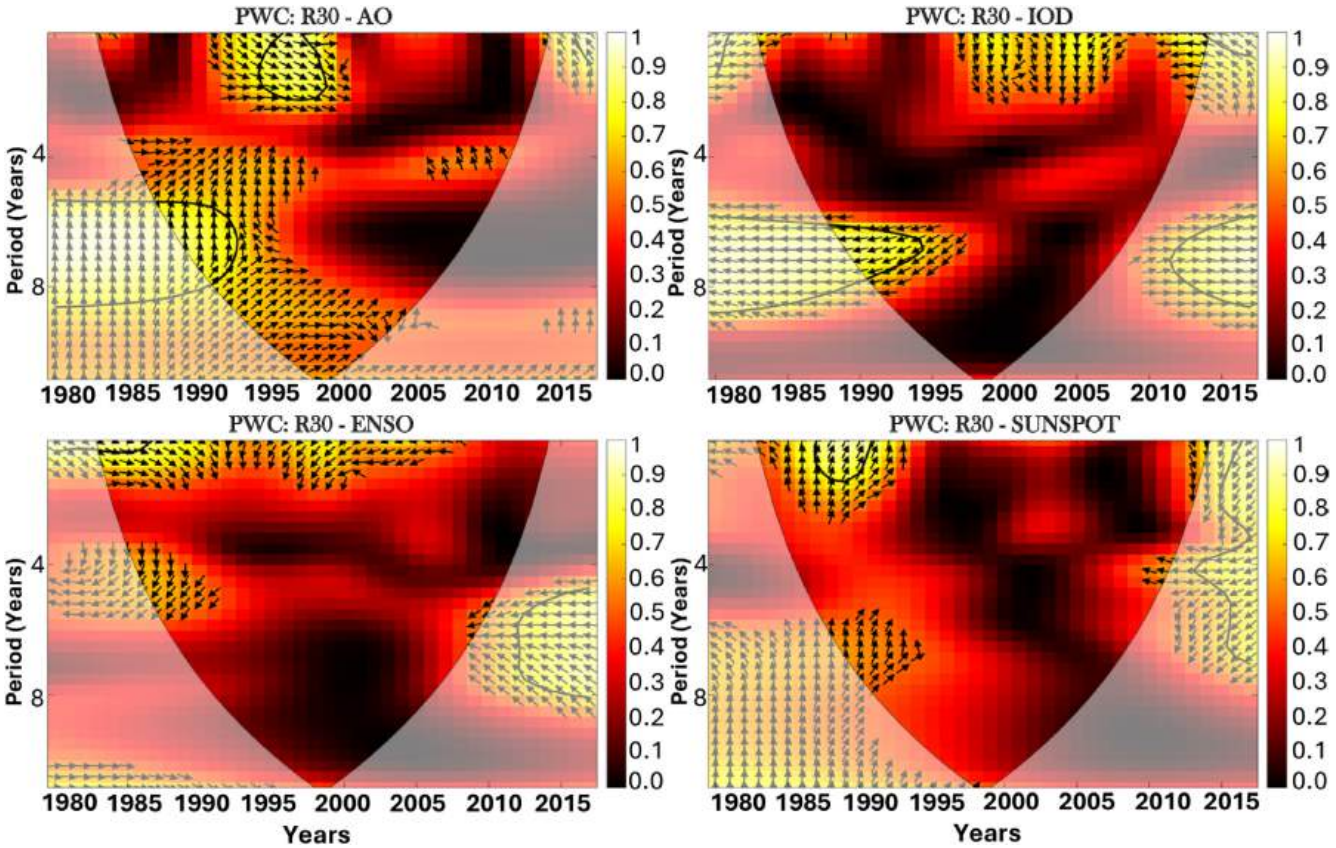

FIGURE 14 | Partial wavelet coherence (PWC) analysis between R30 and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: Rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

bands include 9, 6 and 5 stations, respectively; thus, the strong negative correlations in the 10–30 m class should be interpreted cautiously given the smaller sample size and potential spatial clustering (Rahman and Islam 2019).

4.2 | Trends in Extreme Precipitation Indices in Bangladesh

The EPIs exhibit strong spatial heterogeneity across Bangladesh, reflecting the combined effects of monsoon dynamics, regional physiography and large-scale climate variability. Overall, the north-west shows signatures of weakened monsoonal moisture transport, rising CDD and declining PRCPTOT, whereas the south-east and coastal belt display intensified extremes (higher RX1DAY, RX5DAY and R30), with additional modulation by ENSO and the IOD at interannual scales (Ahmed et al. 2017).

The most coherent signal is the increase in CDD, which has lengthened by ~50 days since the 1980s (≈12.5 days per decade), particularly at inland stations (Rajshahi, Rangpur, Dhaka). This drying is consistent with reduced moisture advection into north-western Bangladesh and may be reinforced by the Meghalaya–Shillong Plateau 'rain shadow' that impedes low-level monsoon inflow (Ahmed et al. 1996). In contrast, coastal districts (e.g., Barisal, Khulna) maintain substantially lower CDD due to direct maritime moisture supply, consistent with reported north-west drought intensification and relatively wetter coastal conditions (Abdullah et al. 2020).

CWD trends are mixed: declines in Dhaka, Sylhet and Mymensingh suggest weaker or less persistent inland convection, while increases at stations such as Comilla, Sitakunda, and Jessore indicate enhanced coastal/south-eastern convection and moisture influx. This coexistence of rising CDD in some regions and rising CWD in others underscores the spatially uneven evolution of monsoon processes (Mishu et al. 2025).

PRCPTOT decreases in north-central and north-western districts (e.g., Rajshahi, Bogra, Faridpur) raise concerns for water availability and agricultural sustainability (Shahid and Khairulmaini 2009; Roxy et al. 2015). Conversely, increases at south-eastern coastal and hilly stations (e.g., Sandwip, Teknaf, Rangamati) are consistent with stronger Bay of Bengal moisture delivery, cyclone-related rainfall, and orographic enhancement in the Chittagong Hill Tracts (Rakib 2019). RX1DAY and RX5DAY display a similar contrast, intensifying in coastal/south-eastern locations (e.g., Cox's Bazar, Sitakunda) but weakening at some inland stations (e.g., Mymensingh), broadly aligning with earlier assessments (Ahmed et al. 2017; Abdullah et al. 2020).

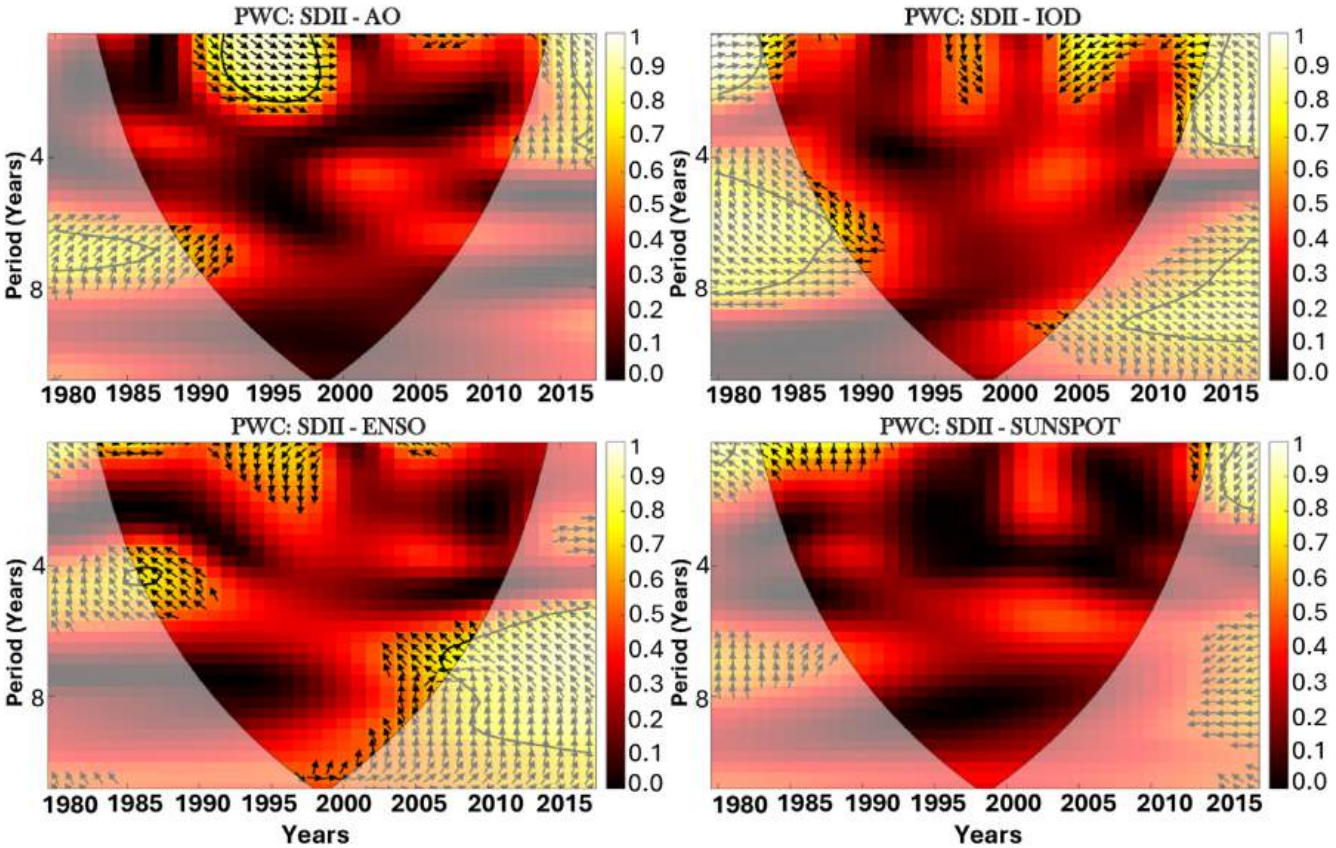

FIGURE 15 | Partial wavelet coherence (PWC) analysis between SDII and large-scale climate drivers (ENSO, PDO, IOD, AMO, AO and SUNSPOT) for 1980–2017. The *x*-axis represents time, while the *y*-axis represents the period (in years). Warmer colours indicate stronger coherence, with black contours enclosing regions significant at the 95% confidence level. The cone of influence indicates areas unaffected by edge effects. Arrows denote phase relationships: Rightward = in-phase, leftward = anti-phase, upward = driver leads, downward = index leads. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 3 | Relationship between meteorological sites' latitude, longitude, altitude, and EPIs.

Index	Latitude	Longitude	Altitude (0–10 m)	(10–30 m)	> 30 m
CDD	**−0.56***	0.26	0.16	−0.55	0.27
CWD	0.03	0.16	0.55	−0.07	−0.74
PRCPTOT	0.22	−0.08	0.15	**−0.73***	0.24
RX1DAY	0.30	−0.15	0.58	0.03	0.98
RX5DAYS	0.12	−0.08	0.50	−0.20	0.89
R10	0.13	−0.16	−0.10	−0.53	0.70
R30	0.16	−0.15	−0.09	−0.70	0.43
SDII	0.13	−0.07	−0.04	**−0.78***	0.75

Note: The symbols *, **, and *** denote significance levels of $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. Bold values indicate significance level.

R10 and R30 also indicate increasing wet-day frequency in the south-east and declining frequencies inland, while Sylhet remains persistently high, consistent with strong orographic uplift and monsoon convergence and associated landslide susceptibility (Islam et al. 2021). SDII reinforces the coastal–inland divergence, increasing at Teknaf and Sitakunda but declining in inland urban centres (e.g., Dhaka, Bogra), where land-use change and urban hydrometeorological feedbacks may complicate local signals (Pandey and Ghosh 2023).

Taken together, the results suggest a dual climatic stressor: progressive drying across north-western/interior regions (increasing CDD and declining PRCPTOT/CWD) and intensifying rainfall extremes along the south-east and coast (increasing RX1DAY,

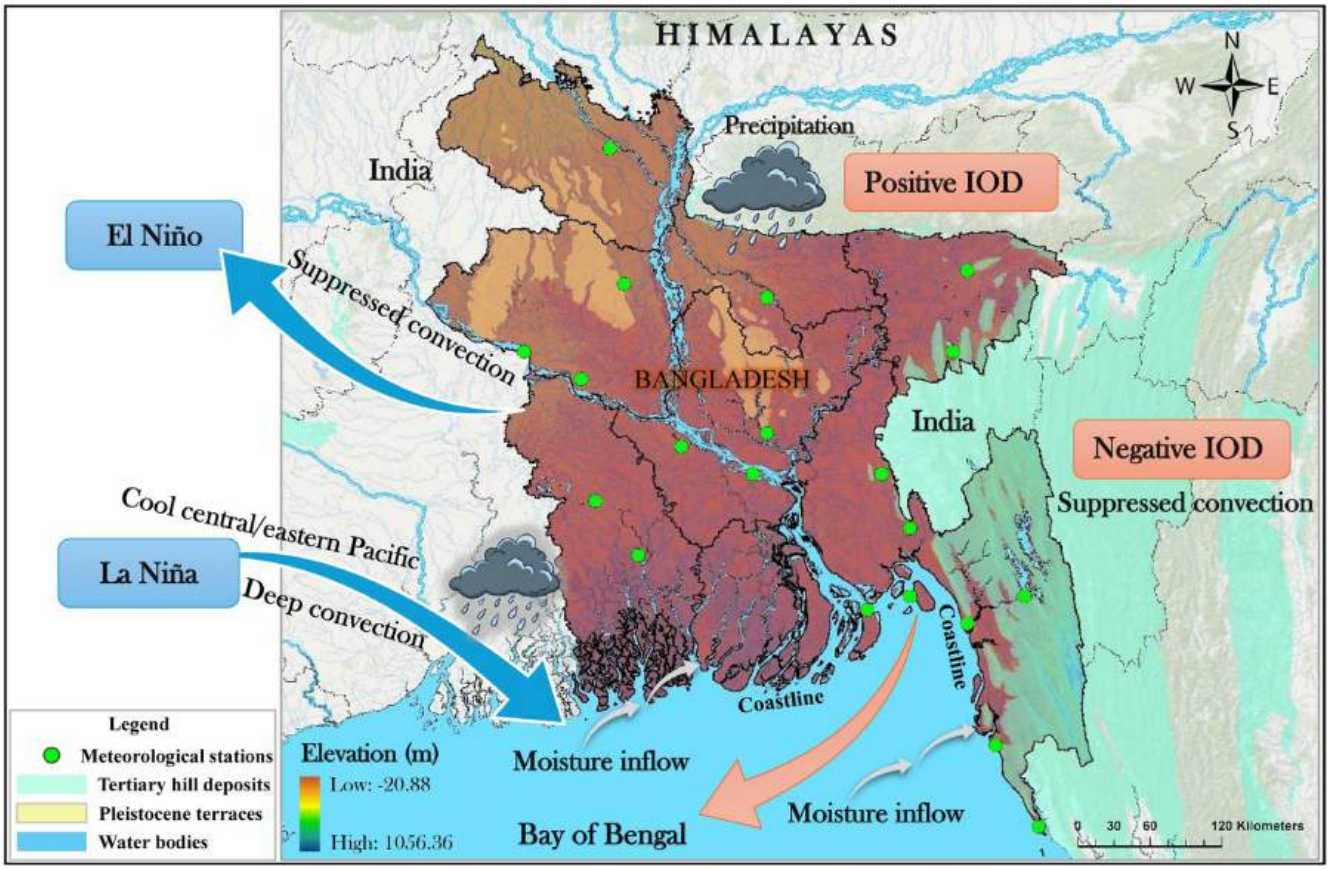

FIGURE 16 | Synergistic effect of ENSO and IOD phases on monsoon-dominated extreme precipitation across Bangladesh. [Colour figure can be viewed at wileyonlinelibrary.com]

RX5DAY, R30 and SDII). This divergence likely reflects the interplay of Bay of Bengal moisture advection, orographic forcing, and circulation anomalies (Kripalani et al. 2003), further modulated by ENSO/IOD variability, with distinct implications for drought management inland and flood/landslide risk reduction in coastal and upland zones.

4.3 | Influence of Climate Drivers on Extreme Precipitation in Bangladesh

Bangladesh receives most of its rainfall during the South Asian summer monsoon (June–September), sustained by strong moisture and heat fluxes from the Bay of Bengal. Large-scale ocean–atmosphere modes—particularly ENSO and the IOD—modulate monsoon rainfall by altering sea-surface temperatures (SSTs), moisture transport, and upper-level circulation, thereby influencing both precipitation frequency and intensity (Rahman and Islam 2019). During El Niño, warming in the central–eastern Pacific suppresses convection over the western Pacific and Bay of Bengal, weakens monsoon south-westerlies, and tends to reduce rainfall over Bangladesh; La Niña exerts the opposite effect, enhancing Bay of Bengal convection and increasing the likelihood of above-normal rainfall and extremes (Ahmed et al. 1996; Ezaz et al. 2022).

Recent evidence across South Asia indicates increasing high-intensity rainfall despite stable or declining seasonal totals, consistent with thermodynamic intensification of the hydrological cycle under warming (Eman et al. 2025). ENSO and the IOD are key drivers of interannual monsoon variability, often with stronger influence during late monsoon months (Krishnamurthy et al. 2009). Bangladesh's location on the northern Bay of Bengal—close to a major centre of deep convection—heightens its sensitivity to Indo-Pacific SST anomalies and associated moisture-flux changes (Kripalani et al. 2003). In general, La Niña and positive IOD conditions favour enhanced moisture convergence over the Bay and increase the likelihood of extreme rainfall over Bangladesh, with the deltaic lowlands and funnel-shaped coastline further amplifying convergence and flood risk (Saha et al. 2021).

The 38-year record (1980–2017) encompasses multiple ENSO and IOD cycles and is sufficient for assessing interannual coherence (2–8 years) in wavelet-based analyses, although non-stationarity in the strength and structure of these modes necessitates careful interpretation, particularly near the cone of influence.

Figure 16 provides a conceptual synthesis of these relationships. Conceptually, LA NIÑA strengthens the Walker circulation and monsoon westerlies, increasing Bay of Bengal moisture influx and promoting heavier rainfall, whereas EL-NIÑO and negative IOD phases tend to suppress convection and favour drought conditions (Rahman and Islam 2019; Roxy et al. 2015). Positive IOD phases—characterised by a warm western and cool eastern

Indian Ocean, shift convection towards the western basin and Bay of Bengal, reinforcing monsoon rainfall; co-occurrence with LA NIÑA can produce synergistic effects and elevate flood risk (Ehsan et al. 2023).

Consistent with these mechanisms, the Mantel test, Geo-Detector (Factor Detector) and PWC analyses indicate that large-scale modes exert substantial control on Bangladesh EPIs, with ENSO and the IOD emerging as dominant modulators, particularly for CDD and RX5day. El Niño-related weakening of Bay of Bengal convection and monsoon inflow is consistent with positive associations with CDD and negative relationships with PRCPTOT and rainfall extremes (Roxy et al. 2015). However, some EL-NIÑO events may enhance regional cyclonic circulation near the Himalayan foothills, increasing intraseasonal variability and permitting short-lived heavy rainfall episodes, which helps reconcile simultaneous suppression of seasonal totals with episodic extremes (Wahiduzzaman and Luo 2021). The IOD influence is likewise physically consistent: positive IOD phases invigorate ascent over the Bay of Bengal and can enhance multi-day extremes such as RX5day, potentially aided by IOD-related changes in upper-ocean stratification and convective potential in the northern Bay (Chen et al. 2022).

Importantly, Geo-Detector provides an objective ranking of drivers, while PWC reveals temporal non-stationarity, with strongest ENSO/IOD coherence at 2–8-year periods concentrated in specific epochs (notably 1990–2005). Overall, these results suggest compensatory dynamics in which positive IOD conditions can partially offset ENSO-induced drying during certain periods, offering an explanation for apparently concurrent associations with RX5day that would be difficult to infer from static correlations alone.

Overall, the findings indicate a dynamic interplay in which ENSO tends to suppress monsoon rainfall and lengthen dry spells, whereas positive IOD phases enhance regional moisture supply and can moderate ENSO-related drying. This underscores the key role of Indo-Pacific climate variability in governing rainfall extremes in Bangladesh, with direct implications for water security, agriculture, and disaster preparedness under continued warming (IPCC 2021).

5 | Conclusion

This study analysed daily precipitation records from 20 meteorological stations across Bangladesh for 1980–2017 to characterise the spatial and temporal behaviour of EPIs, identify years of abrupt change, and assess linkages with major large-scale climate drivers. The main conclusions are as follows:

- Analysis of eight EPIs using the Mann–Kendall test and Sen's slope estimator revealed distinct trends across Bangladesh. CDD exhibited a robust and significant increasing trend, indicating lengthening dry spells, while several wetness-related indices (e.g., PRCPTOT and SDII) weakened at many inland stations. In contrast, short-duration and frequency-based extremes (R10, R30, RX1DAY and RX5DAY) displayed spatially heterogeneous behaviour, with increases at many coastal and south-eastern stations and declines or non-significant trends elsewhere. Sequential Mann–Kendall and Pettitt tests identified notable change points around the late 1980s and late 1990s (e.g., 1988 for CWD and SDII, 1998 for RX5DAY, and 1999 for CDD), after which trends in several indices became more pronounced.
- Spatial analysis showed a clear intensification of extreme precipitation along the south-eastern and coastal belt, particularly around Cox's Bazar and Sandwip, where multi-day extremes and the frequency of heavy rainfall events have increased. By contrast, the north-western region (e.g., Rajshahi) consistently exhibited lower values for most wetness-related EPIs and longer dry spells, indicating increasing aridity. Overall, the results suggest a growing contrast between drying inland areas and increasingly flood-prone coastal and hilly regions.
- The Mantel test revealed statistically significant correlations between EPIs and several large-scale climate modes. In particular, the IOD and ENSO showed strong and recurrent associations with CDD and RX5DAY, underscoring their importance as key modulators of interannual hydro-climatic variability over Bangladesh.
- Geo-Detector (Factor Detector) results identified the IOD as one of the most influential drivers of extreme precipitation, especially for CDD and RX5DAY, with ENSO also contributing substantially. Together, these modes regulate monsoon circulation and moisture transport from the Bay of Bengal, and their impacts may either reinforce or partially offset one another depending on phase combinations.
- Among the drivers examined, Indo-Pacific modes (ENSO and the IOD) exert the strongest and most consistent influence on Bangladesh's extreme precipitation regime. The AO and solar activity show comparatively weak effects.

In summary, the observed changes in extreme precipitation across Bangladesh reflect the combined influence of monsoon dynamics, physiographic controls, and Indo-Pacific climate variability. These findings provide actionable evidence for water-resources management, drought–flood risk assessment, and climate adaptation planning, and they support the development of region-specific strategies to safeguard water security, infrastructure, and agricultural productivity under a changing climate.

Author Contributions

Abu Reza Md. Towfiqul Islam: writing – review and editing, resources, supervision, conceptualization, validation, project administration. **Anika Bushra:** writing – first draft, methodology, analysis and visualization. **Md. Abdullah-Al Mamun:** software, methodology, data analysis and review and editing.

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During the preparation of this work, the author(s) used Grammarly/ChatGPT in order to check the language and clarify the language of the paper. After using this tool, the author(s) reviewed and edited the

content as needed and take full responsibility for the content of the published article.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Autocorrelation of annual precipitation time series for various lags in different stations over Bangladesh. Dash lines represent the significance level at $p < 0.05$. **Table S1:** Station-wise monthly average annual rainfall total (mm), computed from daily observations over 1980–2017. **Table S2:** Detailed information of the correction process for missing records in the daily precipitation data for 20 stations over Bangladesh. **Table S3:** Mann–Kendall trend test results with Benjamini-Hochberg FDR correction. **Table S4:** Geo-Detector method q-statistics with 95% confidence intervals (bootstrap, $n = 10{,}000$). **Table S5:** PWC wavelet coherence bandwidth sensitivity analysis (CDD-ENSO).